University of Pisa
Department of Physics
Master's Degree in Physics of Complex Systems

Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Sources and code: `https://github.com/matteodisante/soaring-anomalous-transport`

Abstract

Soaring birds and human pilots stay aloft by extracting energy from atmospheric updrafts, producing long, intermittent trajectories that alternate climbs within thermals with quasi-ballistic glides. Such motion is a candidate for *anomalous transport*, that is, a departure from ordinary Brownian diffusion. This thesis characterizes the transport statistics of real soaring flights from a large empirical dataset and compares them with stochastic transport models.

This document reports the current state of the project. A first, reproducible acquisition has collected 212,602 flights from the FFVL Coupe Fédérale de Distance across two disciplines: 203,343 paraglider flights (seasons 1999–2000 to 2025–2026) and 9,259 hang-glider flights (2001–2002 to 2025–2026). Of these, 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified. Data collection is ongoing and may be extended to further public soaring databases – notably WeGlide for sailplanes, and XContest – using the same approach. The remainder of the document describes the acquisition method, the dataset and its statistics, and the planned analysis and modelling.

Contents

Chapter 1

Introduction

1.1 Motivation

Gliders, paragliders, and soaring birds carry no continuous source of propulsion; to remain airborne they extract energy from the moving atmosphere, primarily from *thermals*, the buoyant updrafts that form as sunlight heats the ground. A cross-country flight thus proceeds as a repeating cycle of three phases. In the *transition* phase the flyer glides along a nearly straight, energy-efficient path toward the next source of lift. If none is reached at a safe altitude, a *search* phase follows, during which the flyer probes the surrounding air mass to locate lift. Once lift is found, a *climb* phase begins, in which the flyer circles within the rising air to regain altitude. The cycle alternates long, directed displacements with episodes of slow, localized motion.

Intermittent and directionally persistent motion of this kind is the setting in which *anomalous transport* arises. Let $\mathbf{r}(t)$ denote the horizontal position of a flight, measured from take-off. Transport is anomalous when the mean-squared displacement grows as a non-trivial power of time, $\langle|\mathbf{r}(t)|^2\rangle \sim t^\alpha$ with $\alpha \neq 1$, as opposed to the linear growth ($\alpha = 1$) of ordinary Brownian diffusion. Continuous-time random walks and Lévy walks are the standard frameworks for such regimes [1,2], which have been reported in animal movement, in disordered media, and in turbulent flows. Whether the trajectories of real soaring flights belong to an anomalous regime, and of which type, is the question addressed in this thesis.

Soaring has previously been studied mainly as a control problem—how to centre and exploit a thermal efficiently, including by reinforcement learning [3]. The present work is statistical instead: it asks which transport properties emerge at the level of the ensemble of observed trajectories. Its direct precedent is the study of Vilpellet et al. [4], who analysed a large set of human soaring flights, found a horizontal transport that is sub-ballistic and approximately independent of aircraft type (Hurst exponent $H \approx 0.88$), and introduced a phase-resolved analysis based on the transition–search–climb cycle. This thesis builds on that study: it enlarges the dataset and re-examines both the phase segmentation and the transport analysis.

1.2 Objectives

The thesis pursues three objectives:

1. to assemble a large, reproducible, and documented dataset of real soaring flights;

2. to reconstruct clean trajectories from the raw GPS logs and define suitable transport observables;

3. to test for anomalous transport and to compare the empirical statistics with stochastic transport models.

A first dataset—the FFVL CFD archive of paraglider and hang-glider flights—has been assembled and verified and is described here; data collection is ongoing and may incorporate further sources, notably sailplanes (Chapter 2). The second and third objectives define the work ahead (Chapter 4).

1.3 Scope of this document

This document reports the current state of the project and is revised as the work proceeds. Its quantitative content—the summary statistics and tables—is generated directly from the dataset, so it remains consistent with the data in hand. Chapter 2 describes the acquisition of the data and the reproducibility of the pipeline; Chapter 3 describes the dataset, its statistics, and how it is cleaned and characterised into an analysis-ready trajectory; and Chapter 4 sets out the planned transport analysis and modelling that trajectory supports. Throughout, the discussion keeps to the scientific and statistical reasoning behind each choice; implementation-level detail – code modules, exact configuration values, sample sizes, computation times and reproduction commands – is collected separately, at the back of this document, in Implementation and Computational Details (from p. 42) – organised, chapter by chapter, to mirror this body – and reached at the relevant point via a terse cross-reference ("implementation details: Sec. X").

Chapter 2

Data acquisition

2.1 Source: the FFVL Coupe Fédérale de Distance

The data come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition run by the French free-flight federation (FFVL). The federation operates two parallel competitions on separate sites: one for paragliders (`parapente.ffvl.fr`) and one for hang gliders (`delta.ffvl.fr`). Pilots upload their flights; each entry carries metadata (pilot, date, take-off and landing sites, scored distance, duration, glider) and, in most cases, the raw GPS tracklog in IGC format. Both sites expose identical URL structure and machine-readable XML exports, so the same acquisition pipeline handles both. A "season" is identified by its starting year, e.g. the year 1999 denotes the 1999–2000 season; the paraglider archive starts in 1999–2000 and the hang-glider archive in 2001–2002.

2.2 What is collected

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, metadata plus the local path of its track) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository; only the small per-season index is versioned, and it is the source of the statistics in Chapter 3.

2.3 Pipeline and tooling

Acquisition is performed by a dedicated, installable Python package, exposed through two command-line tools, one per discipline, sharing the same subcommands for fetching, downloading, cataloguing, and verifying. The design favours robustness for a long-running, large job: downloads are resumable and safe to interrupt, a response is accepted only once validated as real IGC content, and the client is courteous to the server it downloads from (implementation details: Sec. 2.1).

2.4 Reproducibility

The destination directory is not hard-coded: it is set per machine through an environment variable or a configuration file shipping a placeholder, so no machine-specific path is committed. Every command validates this setting at start-up and aborts with an explicit message if the target is missing, rather than failing obscurely. Archiving the season XMLs verbatim keeps the full provenance: the catalog can always be rebuilt from them (implementation details: Sec. 2.2).

2.5 Integrity

Because the data sit on an external drive, integrity is re-checked independently after download, rather than trusted at download time (implementation details: Sec. 2.3). Both FFVL collections have been verified in full: all 186,052 paraglider and 6,716 hang-glider downloaded tracklogs pass validation (Chapter 3).

2.6 Further data sources

The FFVL CFD is the primary source of the present study, but the dataset is not assumed to be final. The paraglider data (203,343 flights, 186,225 tracklogs) are the main analysis target; the hang-glider data (9,259 flights, 6,746 tracklogs) are collected in parallel and enable the cross-type comparison of Chapter 3. Additional public soaring repositories are being pursued to enlarge and diversify the sample and, above all, to add the third glider category, *sailplanes* (Section 3.3). The most complete sailplane source is *WeGlide*: it exposes flight metadata and IGC tracks through an API, but the default access is rate-limited (of order sixty requests per day), so a bulk export of the archive is not feasible under it; a request for research access with a higher quota has been sent and is pending. Paraglider and hang-glider coverage may likewise be broadened through further portals such as XContest. These sources expose comparable flight metadata and IGC tracks and can be acquired with the same approach and merged into the same catalog.

Chapter 3

The dataset

3.1 The IGC tracklog format

A flight track is stored as an *IGC file*, the plain-text format defined by the FAI/IGC Technical Specification for IGC-approved GNSS flight recorders [5]; it is the standard output of the recorders used in free-flight competitions. Its layout is therefore fixed by that standard, not inferred from the files, and the acquisition validates every downloaded track against it (Chapter 2), so conformance is checked across the whole archive.

Each line is a *record* whose type is set by its first character – a code assigned by the standard, not chosen by the logger. The records relevant here are:

`A` the flight-recorder/manufacturer identifier (first line of the file);

`H` header records (flight date, pilot, glider, geodetic datum, . . .);

`I` the declaration of the optional extra fields appended to every fix;

`B` the GPS fixes – the body of the file.

A `B` record stores its fields at fixed character positions. Decoded against the standard, the example

```
B 094329 4933044N 00005193E A 00095 00149
```

reads as: UTC time `09:43:29`; latitude 49°33.044′ N and longitude 000°05.193′ E on the WGS84 datum; a fix-validity flag; and *two* altitudes in metres – the barometric (pressure) altitude (95 m) followed by the GNSS altitude (149 m).

The validity flag is fixed by the standard to two values and certifies the *GNSS altitude* rather than the horizontal position: `A` marks a valid three-dimensional fix (usable GNSS altitude), whereas `V` marks a two-dimensional fix or a GNSS drop-out, in which case the standard requires the GNSS-altitude field to be written as zero. A `V` fix can thus still carry a usable latitude/longitude and a valid barometric altitude.

Decoding a `B` record from fixed character positions can succeed numerically while the result is not a valid encoding at all – a corrupted byte can still parse as a digit. The parser therefore checks, at decode time and independently of any flight-dynamics judgement, that the record is actually a legal encoding – a valid time, valid latitude/longitude fields, a valid hemisphere letter – dropping it otherwise, on the same footing as a record with the wrong length or a non-numeric field (implementation details: Sec. 3.1). It is a *format*-validity check – is the record even a legal encoding – distinct from the *plausibility* checks of fix-level cleaning (Sec. 3.6.2, e.g. the inter-fix speed bound), which target fixes that decode validly but would be physically implausible given the rest of the trajectory.

The two altitude channels differ in reference and in noise – the barometric value is referenced to the ICAO standard atmosphere and is smooth, the GNSS value is referenced to the WGS84 ellipsoid and is vertically noisier – and on older loggers one channel may be absent (recorded as zero). The analysis adopts the *barometric* channel for the vertical dynamics, for the signal-quality reasons set out and quantified in Sec. 3.6.1; this choice matters because the vertical velocity derived from it drives the phase segmentation.

The sequence of `B` records is thus a time-stamped three-dimensional trajectory. The sampling period is typically a few seconds, is not guaranteed to be uniform, and varies across recording devices – a point the analysis must handle explicitly.

3.2 Flight metadata and the catalog

Every flight in the season XML carries a set of metadata as record attributes, which the acquisition preserves. Beyond the identifiers and links, the fields used downstream are: the **date**; the **pilot** and **club**; the **take-off** and **landing** sites and the French **department**; the glider, given as both a model name (`aile`) and a class (`aile_class`, Section 3.3); and the competition quantities – the **flight type** (the *scoring* category: free distance, out-and-return, flat or FAI triangle, hike-and-fly, . . .), the scored **distance** and **points**, and, where present, the **duration** and average **speed**. The flight type is a scoring label and must not be read as a glider or manoeuvre type.

From these, the pipeline builds a single derived table, the **catalog** (`catalog.csv`), with *one row per flight* across all seasons: the metadata above plus a `local_path` linking each flight to its track on disk and a `downloaded` flag. It is the working index of the project – a flight can be traced to its file, and back (through the identifier embedded in the file name) to its metadata and public page, without any external lookup. A smaller **season index** (`seasons_index.csv`), versioned in the repository, summarises each season by the number of flights declared, carrying a track, and downloaded; it is the source of the statistics below. Both tables are regenerated from the archived XMLs on demand (`build-catalog`).

3.3 Glider categories

Soaring kinematics depend strongly on the aircraft, so the glider must be identifiable for any stratified analysis. Three broad categories exist, in increasing order of performance: **paragliders** (flexible canopy; slowest, glide ratio $\sim$8–11, tight turns), **hang gliders** (rigid-framed delta wing; faster, glide ratio $\sim$10–15), and **sailplanes** (rigid aircraft; fastest, glide ratio $\gtrsim$25). They differ several-fold in cruise speed and turn radius – which is why the reference study [4] reports its transport statistics separately for each.

In practice the category is set by the *data source*, since the public databases are organised by discipline. The FFVL runs the Coupe Fédérale de Distance as two parallel competitions – one for *paragliders* and one for *hang gliders* – so the present dataset spans *both* glider types: the glider-type stratum has two levels, and the cross-type comparison of the reference study [4] is therefore already possible between paragliders and hang gliders. The third category, *sailplanes*, is collected by different portals and is not yet in hand (Chapter 2); until it is added the cross-type analysis is limited to the two FFVL disciplines.

Within each discipline the flights can be further stratified by *performance class*, recorded in `aile_class`; the two disciplines use different class systems. FFVL labels *paragliders* on the EN/AFNOR certification ladder – EN A (“A ou 1”, most stable, recreational), EN B (“B ou 1-2”), EN C (“C ou 2”), EN D (“D ou 2-3”), the “CIVL Competition Class” (CCC) racing wings, plus tandem (“biplace”) and non-certified (“non homologuée”) entries. *Hang gliders* follow the FAI/CIVL class system, which in this data reduces to flexible wings (“Delta Class

1" and the intermediate "Delta Class Sport") and rigid wings ("Rigide Class 2" and "Rigide Class 5"). In both cases the ladder tracks performance – higher classes fly faster and flatter – so it is the natural within-source stratification variable, playing the role that glider type plays across sources.

Two caveats bear on any use of `aile_class`. First, the field must be normalised before it can serve as a stratum, since a small fraction of rows carry a blank or placeholder class. Second, the class certifies the wing, not the pilot. Both points are handled at the pre-processing stage, later in this chapter (implementation details: Sec. 3.3).

3.4 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Chapter 2). It comprises 212,602 flights in total, of which 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified, split across two disciplines.

The *paraglider* collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights, of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 3.1). The *hang-glider* collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6,746 (72.9 %) carry a tracklog and 6,716 (99.6 %) are downloaded and verified (Table 3.2). The hang-glider archive is roughly twenty times smaller, reflecting the smaller active population of the discipline.

In both disciplines the volume of flights grows by orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute thousands of tracks for paragliders and several hundred for hang gliders.

Table 3.1: FFVL CFD *paraglider* flights per season (`parapente.ffvl.fr`): total declared, with a GPS track, and downloaded. (implementation details: Sec. 3.4)

Season	Flights	With GPS	Downloaded	Coverage (%)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0

Season	Flights	With GPS	Downloaded	Coverage (%)
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 3.2: FFVL CFD *hang-glider* flights per season (`delta.ffvl.fr`): total declared, with a GPS track, and downloaded. (implementation details: Sec. 3.4)

Season	Flights	With GPS	Downloaded	Coverage (%)
2001–2002	248	0	0	0.0
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

3.5 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of paraglider entries and about 27 % of hang-glider entries have no IGC file (older or manually declared flights); they carry metadata only.

- **A small residue of unrecoverable files.** A few flights advertise a track that the server does not actually serve as a valid IGC (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, altitude source, and accuracy depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Altitude channel.** The barometric channel is adopted for the vertical dynamics (Sec. 3.6.1), but a sizeable minority of loggers record no pressure altitude – over a quarter of paraglider and roughly a sixth of hang-glider flights – which fall back to the GNSS altitude, flagged as such.
- **Glider metadata.** Flights are paragliders or hang gliders, tagged by source (Section 3.3); the class field `aile_class` uses a different system in each discipline and must be normalised before it is used to stratify.
- **Incomplete dates.** Some historical entries have placeholder dates (e.g. `0000-00-00`); these are kept but flagged.

These caveats motivate the pre-processing described in the remainder of this chapter.

3.6 Trajectory pre-processing

Each IGC file is parsed into a sequence of fixes $(t_i, \varphi_i, \lambda_i, h_i)$ – UTC time, geodetic latitude and longitude, and altitude – by the `soaring.analysis.igc` parser, which reads the `B` records at their fixed character positions and returns both altitude channels. The pipeline then proceeds in a fixed order: (i) choose the altitude channel; (ii) clean individual fixes; (iii) trim the ground phases; (iv) filter out unfit whole flights; (v) convert to a local Cartesian frame; (vi) enforce a uniform sampling interval within each flight; and (vii) smooth and differentiate the trajectory. The output is a tidy, analysis-ready representation in a growing `soaring.analysis` sub-package.

The ordering follows one rule: work in the frame that makes each step exact and cheap. Steps (i)–(iv) act on the *raw geographic* coordinates, since the only geometric quantities they need – inter-fix speed and spatial extent – follow from the great-circle (haversine) distance between fixes, with no coordinate conversion. The conversion (v) to the metric East–North–Up frame (Sec. 3.6.5) is then done on the cleaned, trimmed track, with the origin at the take-off fix ($\mathbf{r}(0) = \mathbf{0}$), so the tangent-plane frame is built from good data and referred to the right origin. The final steps (vi)–(vii), which produce filtered position, velocity and acceleration, are naturally metric and so run *after* the conversion.

3.6.1 Choice of altitude channel

The vertical dynamics – vertical velocity, climb/sink, and the phase segmentation that rests on them – are taken from the *barometric* altitude rather than the GNSS altitude. The reason is one of signal quality in the frequency band that matters. The barometric (pressure) altitude has sub-metre relative precision and very little high-frequency noise; its errors are slow – a constant offset from the ICAO reference and a gradual drift with the synoptic pressure over a multi-hour flight – and slow errors are annihilated by the differencing that every dynamical quantity involves (a vertical velocity, a displacement increment, an autocorrelation). The GNSS altitude, by contrast, is geometrically referenced but vertically noisier: the vertical dilution of precision makes it 1.5–2 times worse than the horizontal, and multipath adds high-frequency jitter of a few metres – exactly the band that differencing *amplifies*. This is also why the variometers used in free flight derive the climb rate from the pressure sensor, not from GNSS.

Empirical confirmation. This is borne out on our data, with one qualification that turns out to matter. Figure 3.1 compares the two channels over a subsample sized for a robust spectral estimate (Appendix B) (implementation details: Sec. 3.5.1). At low frequency the two agree, tracking the real climbs and glides. The high-frequency behaviour is where they part, but *not uniformly*. For the *typical* flight the two are comparable: both approach the common floor set by the metre resolution to which the IGC format rounds *both* channels, so the median spectra (panel c, solid lines) nearly coincide. What separates the channels is the *spread* across flights, the shaded bands. The barometric band stays tight – that channel is uniformly clean – whereas the GNSS band fans out by one to two orders of magnitude towards the Nyquist frequency: a substantial minority of flights carry large GNSS high-frequency noise, from multipath and the vertical dilution of precision, while most do not. Since the vertical velocity is a finite difference, which amplifies exactly that band, and since one cannot know in advance which flights have poor GNSS reception, the uniformly-clean barometric channel is the safe choice – the same reasoning by which free-flight variometers read the climb rate from the pressure sensor, not from GNSS. This *departs from the reference study* [4], whose pipeline derives the vertical coordinate and velocity from the GNSS altitude; the divergence is deliberate, and one of the points re-examined here.

Several practical points follow.

One channel per flight. A trajectory uses a single altitude channel end-to-end; the barometric and GNSS values are never spliced within a flight, since a mid-flight switch between two channels offset by tens of metres would inject a spurious jump.

Source recorded. Which channel a flight uses is stored as a per-flight field – an `alt_source` column (`baro` or `gnss`) in the processed dataset, alongside duration, fix count and the like; the raw IGC files are never modified.

Availability. A flight is used with its barometric channel when it has one, and the few individual fixes with a missing barometric value are dropped rather than back-filled from GNSS. A logger with *no* pressure sensor writes the whole barometric channel as zero (the presence is essentially all-or-nothing, not a scattering of gaps); such a flight falls back to the GNSS altitude, unfiltered – a non-negligible minority of flights (Figure 3.1d).

Measuring the fallback rate. Because it is a headline number, its precision is exact rather than estimated: both archives are full-censused – 17.5 % of the 6,716 hang-glider and 30.1 % of the 186,025 paraglider flights scanned carry no barometric channel (implementation details: Sec. 3.5.1). (The counts quote the census scan directly, through generated macros; the archive keeps growing by a trickle, so they can trail the live download totals of Sec. 3.4 slightly until the scan is refreshed.)

Is a mixed dataset a problem? Not much, for two reasons. The *horizontal* transport analysis – the primary result (MSD and its scaling) – does not use the altitude at all, so every flight is usable there regardless of channel. Altitude enters only the vertical velocity, hence the segmentation and the phase-resolved observables; there the two source groups are kept distinct through `alt_source` and their statistics checked for compatibility (Sec. 3.7). If the noisier GNSS-only flights were to bias a vertical statistic, that analysis is restricted to the barometric flights – still the large majority – while the horizontal analyses keep the full sample. Forcing GNSS on *all* flights, to avoid the mix, would be the wrong trade: it would needlessly degrade the vertical signal of the barometric majority to match the worst loggers.

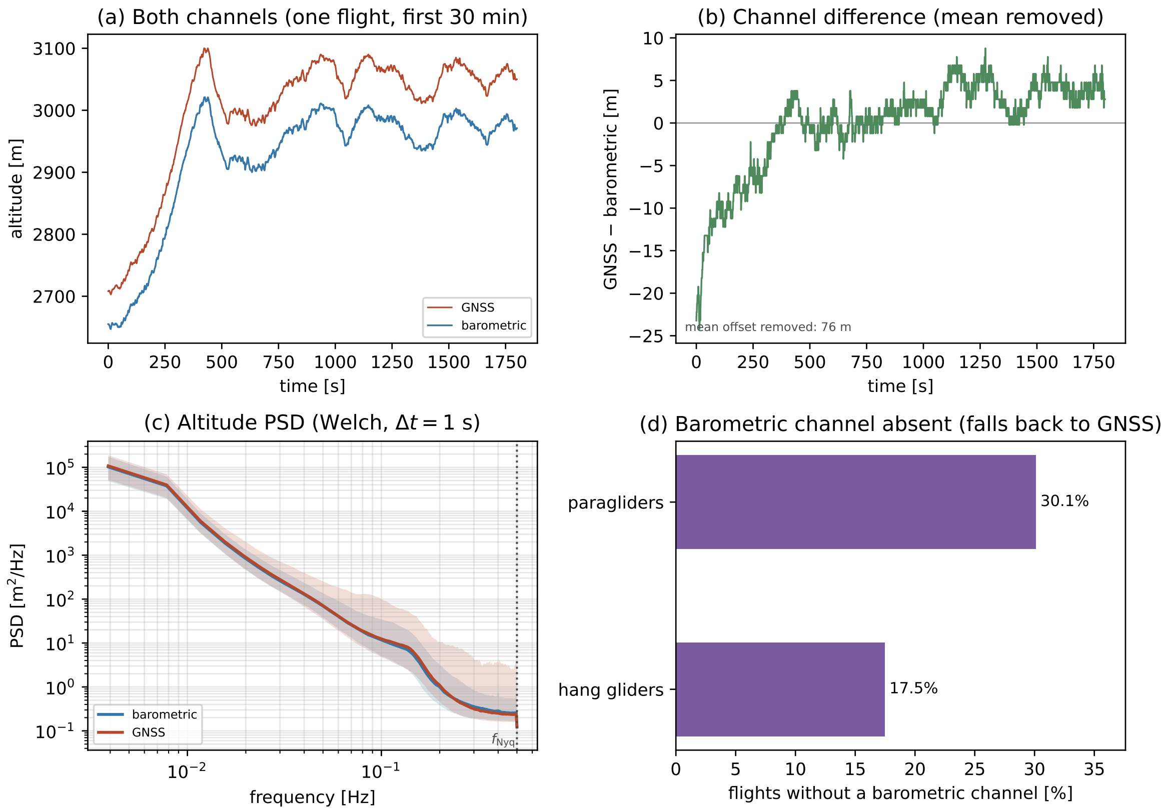

Figure 3.1: Barometric vs GNSS altitude noise. Panels (a)/(b): one representative flight over a 30 min window. (a) the two raw channels on a single *shared* axis, so their offset – the gap between the pressure and ellipsoidal references, tens of metres – is read directly, while both track the same vertical motion. (b) their difference GNSS−barometric (mean removed, *not* detrended): the offset drifts by tens of metres over the flight, a real feature – plausibly the barometric channel's slow departure from the ICAO standard atmosphere as altitude changes – not noise, so only its constant part (annotated) is subtracted. Panels (c)/(d): over the ensemble. (c) the per-frequency *median* Welch spectrum of each channel (Appendix B), pooled over the two disciplines (whose spectra coincide), with a shaded 10 to 90 percentile band across flights. The median lines nearly coincide: for the *typical* flight the two channels are comparable at high frequency, both near the floor set by the shared metre-resolution logging. The *bands* carry the real story – the barometric band stays tight (uniformly clean), whereas the GNSS band fans out by one to two orders of magnitude towards Nyquist, driven by a minority of flights with heavy multipath noise. The median, not the mean, precisely because these per-flight spectra are heavy-tailed and a mean is dominated by that noisy minority (Sec. 3.5.1). The dotted line is the Nyquist frequency $f_{\mathrm{Nyq}} = 1/(2\Delta t)$. (d) Fraction of flights whose barometric channel is absent, per discipline. (implementation details: Sec. 3.5.1)

3.6.2 Fix-level cleaning

Geometry on the raw coordinates. Cleaning, trimming and flight-level filtering all run *before* the Cartesian conversion (Sec. 3.6.5), directly on the geodetic angles (φ_i, λ_i). Distances between fixes therefore cannot be Euclidean; they are great-circle (haversine) distances on a sphere of mean radius $R_\oplus = 6{,}371\,\mathrm{km}$. For two consecutive fixes,

$$\Delta s_i = 2R_\oplus \arcsin\sqrt{\sin^2\frac{\varphi_{i+1}-\varphi_i}{2} + \cos\varphi_i\cos\varphi_{i+1}\,\sin^2\frac{\lambda_{i+1}-\lambda_i}{2}}\,, \tag{3.1}$$

from which the horizontal speed between the fixes is $v_{xy,i} = \Delta s_i/(t_{i+1}-t_i)$ and, from the barometric altitude, the vertical speed is $v_{z,i} = (h_{i+1}-h_i)/(t_{i+1}-t_i)$. Two whole-flight quantities used below follow from the same distance: the flown *path length* $L = \sum_i \Delta s_i$ and the *extent* $D = \max_i \Delta s(\mathrm{fix}_0, \mathrm{fix}_i)$, the farthest a fix gets from take-off. The spherical approximation is ample here – the ellipsoidal correction to an inter-fix distance is $\lesssim 0.3\,\%$, far below the GPS noise – and it avoids converting coordinates before the trajectory has been cleaned.

One rule: clean without biasing the transport. Individual fixes that are corrupt or physically impossible must be removed without discarding the flight around them. But every fix removed or altered is an intervention on the trajectory whose statistics the analysis is about to measure, and an over-eager cleaning biases the result as surely as dirty data does. The design rests on a single rule: assume as little as possible about the motion. The only assumption used is smoothness at the scale of the sampling – an aircraft cannot jump between fixes a few seconds apart – never a model of the transport itself. A model-based smoother would reject outliers more aggressively, but at an unacceptable price: fitting a constant-velocity or diffusive model to the track would write the behaviour under study into the cleaned data.

The two errors this rule guards against are not symmetric, and the cleaning is deliberately asymmetric in response. The slow, localized phases of a flight – search and climb – are precisely the *waiting times* of the transport model (Ch. 4), and their durations populate the tail of $\psi(\tau)$ where the anomalous exponent is decided. Leaving a spurious fix in a slow phase has a bounded, recoverable effect – the smoother absorbs it, an absolute bound catches it – whereas *removing* a genuine slow-phase fix erases a real trapping event and biases that tail irreversibly. The cleaning therefore defaults to *keep*: a fix is removed only on positive, corroborated evidence of a defect, never on suspicion, and no detector may delete a fix that still carries a usable horizontal position (it may at most mark a channel missing, to be restored downstream). This is what lets the procedure strip the plainly broken without ever eating a search or climb fix.

Three kinds of defect, three tests. Past the format-validity check already made at decode time (Sec. 3.1), which rejects any record that is not a legal encoding, a surviving fix can still be wrong in one of three ways, each caught differently (Table 3.3). A fix *off the trend* – a GPS position spike, or a jump out and back within one step – is found by a robust local-outlier test: its residual r_k from a median estimate built on the fixes around it is scaled by the robust local spread of those residuals, and the fix is flagged when

$$r_k > \max\big(k\,\sigma_k,\ \varepsilon_{\min}\big), \qquad \sigma_k = 1.4826\,\mathrm{MAD}_k, \tag{3.2}$$

i.e. when it departs by more than k robust standard deviations *and* by more than an absolute floor $\varepsilon_{\min}$ of a few times the GPS noise (parameters and rationale in Sec. 3.5.2); an absolute speed bound stays underneath as a backstop. A fix *on the trend* agrees with its neighbours yet is still wrong as a record – a duplicated or backward timestamp, or a run of frozen positions from a lost GNSS lock; its residual is near zero, so it escapes the outlier test and needs a structural rule instead. A value *out of range*, an impossible altitude or climb rate, is caught by an absolute bound that needs no context. (Status: the decode-time format check is implemented and unit-tested, and the absolute bounds are thresholded on the data; the detectors themselves are designed but the routine that walks a trajectory applying them is still to be built – Sec. 3.5.2 tracks this precisely.)

Which fix is removed, and what happens to it. This is where a raw speed bound falls short: a speed belongs to the segment *between* two fixes and cannot say which endpoint is at fault. The local-outlier test settles it by asking, of each fix alone, whether it strays from what its neighbours predict, so the culprit is named directly – a lone spike is removed without dragging a good neighbour with it, and a short burst is taken out as one block, the smallest whose removal restores the trend. What follows depends on the fault. A corrupt *position or time* deletes the fix as a node – it is removed, *not* overwritten with the local median: detection is the outlier test's only job, and reconstruction is left to the single resampling step, so the two never mix. It leaves the sequence, its neighbours close up, and the small gap is bridged later, at resampling (Sec. 3.6.6). A corrupt *altitude* keeps the position and time and marks only the altitude missing; since the horizontal analysis – the primary result – never uses altitude, clearing a barometric spike costs it nothing (Sec. 3.6.1).

The frozen-lock case, and why it never catches a climb. One structural defect strikes the observable itself. When the GNSS loses lock, many loggers repeat the last position for tens of seconds while the clock runs on: horizontal speed zero, altitude in range, timestamps valid, every bound above satisfied. Read into the flight as a long *waiting time* such a run would plant a false trapping event in the tail of $\psi(\tau)$, where the anomalous exponent is most fragile; but here the *opposite* error is the one to fear, because a tight thermalling climb is also slow and localized, and wrongly splitting it would *destroy* a real trapping event. The detector is therefore built to fire only on positive, corroborated evidence of a lost lock and to keep the data otherwise. Three independent signatures separate a true freeze from a real climb, and a run is cut only when they agree:

(i) *Spatial extent, not step.* A lost lock collapses the whole run into the logger's quantization: *every* fix of the run lies within ε (a few times the GPS noise) of every other, so the run's bounding diameter stays below ε. A circling climb instead traces a disc of diameter $2R \approx 40\,\mathrm{m}$ to $100\,\mathrm{m}$, far larger than ε, and clears the test on extent alone. The criterion is the run's *extent*, never the fix-to-fix step: at 1 Hz a climbing wing advances $\sim 10\,\mathrm{m}$ per fix, comparable to ε, so a step-based test would flag every climb.

(ii) *An independent witness – the barometric altitude.* A GNSS lock-loss freezes only the GNSS position; the pressure sensor is a *separate* instrument. A wing that is really thermalling keeps gaining height on the barometric channel – a climb rate of order $1\,\mathrm{m\,s^{-1}}$ over the run is tens of metres – whereas a true freeze shows a flat barometric trace. A horizontally motionless run whose barometric altitude is still changing at a plausible rate is a (tight) climb, and is *kept.*

(iii) *The recorder's own flag.* A lost lock is frequently self-declared – the `V` validity flag, the GNSS altitude written as `00000` (Sec. 3.1) – and repeats byte-identical coordinates (zero displacement, not merely small). Both corroborate a freeze; neither occurs in real motion.

The rule, in one line: a run is read as a loss of signal only when

$$\underbrace{\mathrm{diam(run)} < \varepsilon}_{\text{(i) collapsed}} \wedge \Big(\underbrace{|\Delta z_{\mathrm{baro}}| < \delta_z}_{\text{(ii) baro flat}} \vee \underbrace{\texttt{V}/\text{zero-alt/byte-identical}}_{\text{(iii) declared}}\Big) \wedge \underbrace{\Delta t_{\mathrm{run}} \geq \tau_{\mathrm{freeze}}}_{\text{duration}}, \qquad (3.3)$$

i.e. spatially collapsed *and* (barometrically flat *or* explicitly declared), lasting at least τ_{freeze}: it is then marked as a gap and the trajectory is *split* there rather than interpolated across, inventing no motion over an interval whose true path is unknown (Sec. 3.6.6). One corner case falls out of (3.3) correctly: a run whose coordinates repeat *byte-identically* while the barometric altitude climbs is cut, not kept – no real circling wing rewrites the same coordinates, so the positions are invented even where the height gain is real, and splitting there loses only the wind drift the receiver never recorded. The sole residual blind spot – a GNSS-fallback flight (no barometric witness) circling so tightly that its diameter drops below ε, and with no `V` flag – is a near-empty set, and there the default to keep errs on the safe side, preserving a rare true freeze rather than risking a real climb.

Defect	Detected by	Action
Off the trend – corrupt position		
Position spike, out-and-back jump	robust local-outlier test; absolute v_{xy} bound ($45/55\,\mathrm{m\,s^{-1}}$, para/hang)	delete fix, rejoin neighbours
On the trend – structural		
Duplicate timestamp	two or more fixes in one UTC second	merge to one (centroid)
Backward timestamp	time decreases, and not a midnight rollover	delete fix
Frozen-lock run	run extent $< \varepsilon$ *and* baro. altitude flat (or `V`/zero-alt flag, byte-identical repeats), span $\geq \tau_{\text{freeze}}$	mark as gap, split there
Out of range – corrupt altitude, position kept		
Missing altitude	adopted channel absent at the fix	drop altitude only
Altitude out of band	outside $[-100, 6000]\,\mathrm{m}$	drop altitude only
Vertical-speed spike	barometric $\lvert v_z \rvert > 13\,\mathrm{m\,s^{-1}}$	drop altitude only

Table 3.3: Fix-level cleaning, grouped by how each defect presents. *Off the trend*: an isolated fix its neighbours do not predict; the robust local-outlier test (Sec. 3.5.2) names it, and the absolute horizontal-speed bound is a per-discipline backstop – paragliders and hang gliders have different speed envelopes (Fig. 3.2), so one shared value would either clip the faster type or spare the slower one; a `sailplanes` entry follows once that source is in. *On the trend*: defects consistent with the local motion, invisible to the outlier test, each with a structural rule; the frozen-lock run (its extent tolerance ε, minimum span τ_{freeze} and independent barometric witness in Sec. 3.5.2) must satisfy Eq. (3.3) – collapsed, *and* flat or declared, *and* long enough – before it is cut, so a tight climb is never split, and is then split, not bridged, so no motion is invented (Sec. 3.6.6). *Out of range*: absolute sensor bounds on the adopted altitude channel (Sec. 3.6.1); the lower bound is $-100\,\mathrm{m}$, not 0, because the barometric channel is not QNH-corrected and its offset (Fig. 3.1, of order $100\,\mathrm{m}$) lets a sea-level take-off read slightly negative. A corrupt altitude never discards the fix's horizontal position. Inter-fix time gaps are not a criterion here: a gap is a property of the sampling, handled once at the flight level (Sec. 3.6.6). (implementation details: Sec. 3.5.2)

The validity flag needs no rule of its own. The IGC validity flag (`A`/`V`) certifies the GNSS altitude only, so its effect is already covered by the missing-altitude rule on the adopted channel. A `V` fix keeps a valid position and, where present, a valid barometric altitude (Sec. 3.1): on a barometric flight it is kept, on a GNSS-fallback flight it has no usable altitude and drops out as missing. Neither case needs handling of its own.

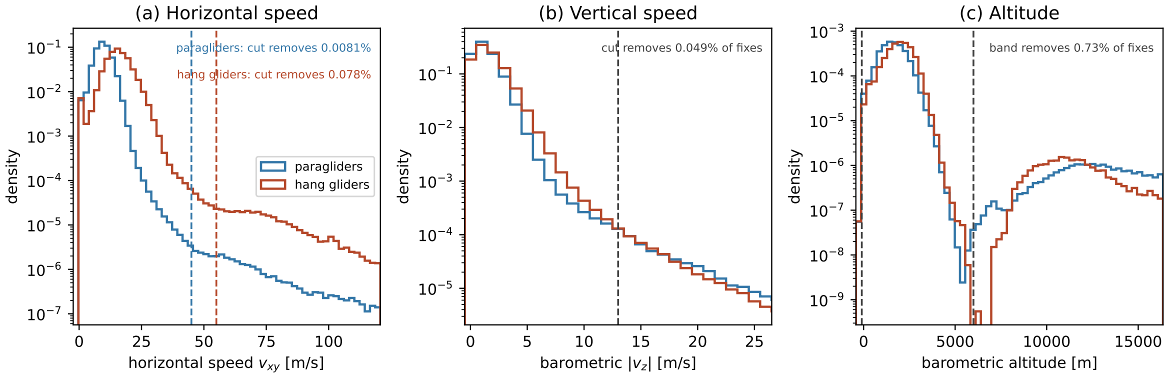

Figure 3.2: Fix-level cleaning: auditing the absolute bounds. Per-*fix* distributions of the quantities the out-of-range bounds test – (a) horizontal speed v_{xy}, (b) barometric $|v_z|$, (c) barometric altitude – pooled over a large paraglider subsample and the complete hang-glider population, with the adopted bounds dashed (Table 3.3); the y-axis is logarithmic so the sparse tail the bounds act on is visible. These are the *context-free* cuts, the floor on the plainly impossible; the robust local-outlier and frozen-lock tests act on structure a marginal distribution cannot show and are validated separately (Sec. 3.5.2). Panel (a) is why the speed bound is per-discipline: hang gliders show a distinct, non-decaying plateau from about $55\,\mathrm{m\,s^{-1}}$ to $90\,\mathrm{m\,s^{-1}}$ on top of the steeply decaying real distribution – a sustained *ground* speed of $200\,\mathrm{km\,h^{-1}}$ to $320\,\mathrm{km\,h^{-1}}$ has no airborne explanation for a hang glider and reads as a logger artefact; the bound is placed where the decay *stops*, not where it first slows, since a shallow but real decreasing stretch can still be genuine rare dynamics. Paragliders show the same pattern lower down. Panel (b) uses one bin per integer $\mathrm{m\,s^{-1}}$: barometric altitude is logged to the metre, so a finite difference returns integer rates and a finer grid would alias them into a spurious comb; its $13\,\mathrm{m\,s^{-1}}$ bound sits above the strongest thermals and ordinary sink but below what a sustained spiral dive can reach, so the validation checks that the fixes it cuts are isolated, not run-shaped (Sec. 3.5.2). Panel (c) uses a narrower range, its cut sitting in an otherwise tight, unimodal distribution. (implementation details: Sec. 3.5.2)

Every fix keeps a full (x, y, z) **– eventually.** The segmentation (Ch. 4) needs a defined horizontal position *and* altitude at every time, since the vertical velocity is its discriminant. Cleaning does not guarantee this fix by fix – it may delete a node (a corrupt position) or mark a channel missing (a barometric spike, a `V` fix on a GNSS-fallback flight) – because the guarantee is established one step later, at resampling (Sec. 3.6.6): within each retained segment every grid time is given a complete (x, y, z), the short holes filled by interpolation per channel and the long ones removed by splitting. The invariant is thus "complete (x, y, z) *within* a segment", not across a split. This is why the validity flag, and the missing-altitude rule it feeds, need no repair of their own here: a dropped altitude is not a lost altitude but a *deferred* one, reconstructed downstream where a single, audited interpolation does all the filling.

A flight-level backstop. Cleaning acts fix by fix, but it also keeps count of how much it had to do. If it has removed or reconstructed more than a small fraction of a flight's fixes, the flight is no longer being cleaned but rebuilt, and it is dropped whole – the cleaning counterpart of the sampling-regularity cut (Sec. 3.6.6).

Validating the cleaning. That each bound removes only a negligible fraction of fixes shows the cleaning is not too aggressive; it says nothing about what the cleaning *misses*. Two checks address that, and both belong to the planned implementation (Ch. 4). *Injected defects*: spikes, frozen runs and time glitches of known size are planted in clean tracks, and the fraction recovered as a function of size measures the false-negative rate directly. *Invariance of the result*: the transport estimators – the displacement-scaling exponent and the waiting-time tail – are recomputed as the cleaning is tightened, and the operating point is taken on the plateau where they stop moving. A cut that removes error and not signal leaves the physics unchanged; one that has begun to bite into real dynamics does not (implementation details: Sec. 3.5.2).

3.6.3 Trimming of ground phases

The logger starts recording before take-off and stops after landing; these ground phases must be stripped before any kinematic analysis. Detection uses the *horizontal* speed v_{xy} alone (from (3.1)) – the vertical speed plays no role here: the airborne segment begins at the *first* time v_{xy} rises above $5\,\mathrm{m\,s^{-1}}$ and stays above it for at least $30\,\mathrm{s}$, and ends where the *last* such sustained-above stretch ends – the same rule applied to the time-reversed track, *not* a cut at the first sustained drop, which a slow soaring phase (ground speed near zero when hovering into wind) would otherwise trigger mid-flight. The sustained-speed requirement prevents a gust or a GPS glitch on the ground from being mistaken for take-off. One caveat is structural: the criterion is a *ground* speed, so a slow phase adjacent to take-off or landing (a launch straight into ridge lift, say) can still be clipped; the trimmed fraction is recorded per flight so this failure mode is auditable rather than hidden. Both the threshold and the persistence affect what fraction of each trajectory is retained, so they are made explicit (implementation details: Sec. 3.5.3).

3.6.4 Flight-level filtering

After cleaning and trimming, some flights are still unfit for the ensemble analysis and are dropped whole. The step is delicate: too aggressive a cut biases the transport statistics – a high minimum duration, for instance, preferentially keeps successful long flights and inflates the apparent persistence of the MSD. The flight-level filtering therefore uses a **minimal-threshold strategy**: for each criterion, the threshold is set to the smallest value that still clears non-genuine flights (sled runs, tows, aborted or truncated logs), so the ensemble is disturbed as little as possible. Once chosen, each cut is verified by the full-distribution diagnostics in Fig. 3.3: it sits in the empty tail of its distribution, where moving it changes almost nothing. All flight-level quantities are computed directly from the parsed tracks; the thresholds follow the minimal-threshold strategy just described. Two track quantities gate a flight, plus a metadata stratifier:

(1) *Recorded duration* $T \geq 40\,\mathrm{min}$. A cross-country flight spans many climb–glide cycles; shorter logs are sled runs or aborted attempts. The cut is deliberately *decoupled* from the maximum MSD lag – the ensemble at lag t already uses only flights with $T > t$, and the aging in T is diagnosed by stratifying in T – so large-lag analyses restrict to the long-duration stratum rather than raising this global cut.

(2) *Flown path length* $L \geq 30\,\mathrm{km}$, with $L = \sum_i \Delta s_i$ from the great-circle steps of (3.1). This removes tows and local hops that never became a cross-country flight. It is a minimal cut: a real XC flight covers many tens to hundreds of kilometres, far above $30\,\mathrm{km}$.

(3) *Glider class* (metadata). For class-stratified analyses, flights whose `aile_class` cannot be normalised onto the discipline's system are dropped from the class-resolved sub-analyses but kept in the pooled one. Stratification by glider *type* (paraglider vs hang glider) is already possible; the sailplane stratum follows once that source is in hand.

A separate minimum-fix-count cut would be redundant: at $T \geq 40$ min even a 10 s logger – the slowest common hang-glider rate (Fig. 3.7) – already records $\sim$240 fixes, ample for the kinematics; the thin tail of still slower loggers is handled by the Δt stratification (Sec. 3.6.6), not by a fix-count cut. A further criterion – sampling regularity – is applied later, where a flight's time base is examined (Sec. 3.6.6) (implementation details: Sec. 3.5.4).

Figure 3.3 shows, for every discipline and computed on the *full* dataset (every track, no sub-sampling), the distributions of the two track cuts and, beside each, the fraction of flights that cut *alone* would retain – its marginal effect, not a cascade. Both cuts sit far out in the tail, so the retained fraction is insensitive to their exact value; the diagnostics must be re-examined for the less curated further sources.

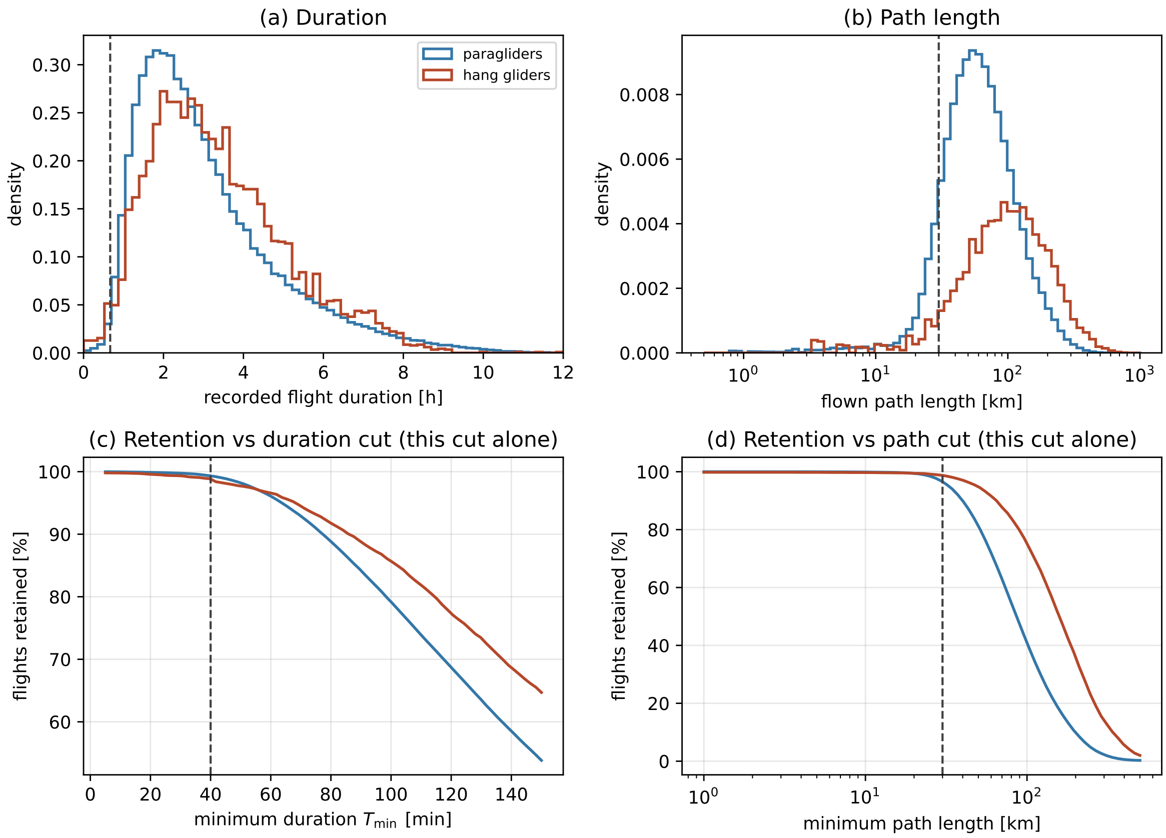

Figure 3.3: Flight-level filtering diagnostics, computed on the *full* dataset – every parsed track, per discipline (paragliders and hang gliders; sailplanes later) – not from the declared metadata and not sub-sampled. Top: distributions of (a) recorded flight duration and (b) flown path length. Bottom: the fraction of flights retained versus (c) the duration cut and (d) the path-length cut, each computed for *that cut alone* (marginal, not cascaded). Dashed lines mark the adopted cuts ($T \geq 40$ min, $L \geq 30$ km), which sit far out in the tail. (implementation details: Sec. 3.5.4)

3.6.5 From geographic to local Cartesian coordinates

The cleaned, trimmed track is next mapped from geographic to local Cartesian coordinates; the two remaining steps – resampling and smoothing – are naturally metric and run on the result. Latitudes and longitudes are angles on a curved surface and cannot be differenced directly to obtain distances; the mapping proceeds in two steps, following the convention of Vilpellet et al. First, each fix is converted to Earth-Centred, Earth-Fixed (ECEF) coordinates on the WGS84 ellipsoid (semi-major axis $a = 6{,}378{,}137\,\mathrm{m}$, flattening $f = 1/298.257223563$, $e^2 = f(2-f)$) [6, 7],

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \qquad \begin{aligned} X &= (N + h) \cos\varphi \cos\lambda, \\ Y &= (N + h) \cos\varphi \sin\lambda, \\ Z &= \big(N(1 - e^2) + h\big) \sin\varphi, \end{aligned} \tag{3.4}$$

where $N(\varphi)$ is the prime-vertical radius of curvature and h the ellipsoidal height, measured along the ellipsoid normal.

A point of care: the latitude φ in (3.4) is the *geodetic* latitude – the angle between the equatorial plane and the ellipsoid *normal* at the point (Figure 3.4), *not* the geocentric latitude ψ, which is the angle to the line from the Earth's centre O. On an ellipsoid the two differ, because the normal does not pass through O (Figure 3.4): the gap $\varphi - \psi$ reaches $\approx 0.19°$ (about $11'$) near $45°$, which if the two were confused would misplace a fix by up to $\sim$20 km. We use the geodetic latitude throughout, and no conversion is needed: WGS84 – hence the GPS receiver and the IGC `B` records (Sec. 3.1) – reports geodetic latitude, and (3.4) with $N(\varphi)$ is precisely the geodetic-to-ECEF transform that consumes it. The longitude λ carries no such ambiguity: it is the same angle in both conventions, since the ellipsoid is a surface of revolution about the polar axis. The geodetic latitude and longitude are exactly the (φ, λ) that orient the local frame below.

Second, taking the first fix $(\varphi_0, \lambda_0, h_0)$ of the flight – the take-off fix, since the ground phase has already been trimmed – as the origin, the ECEF displacement $\Delta\mathbf{X} = \mathbf{X} - \mathbf{X}_0$ is rotated into the local East–North–Up (ENU) frame tangent to the ellipsoid at that point,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0 \cos\lambda_0 & -\sin\varphi_0 \sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0 \cos\lambda_0 & \cos\varphi_0 \sin\lambda_0 & \sin\varphi_0 \end{pmatrix} \Delta\mathbf{X}. \tag{3.5}$$

The ENU frame *is* the local Cartesian frame: its horizontal plane (E, N) is the "north–east" plane and U is the local vertical, so there are not two distinct coordinate systems but one, reached through the intermediate (global) ECEF representation. Over the spatial extent of a single flight (tens of kilometres) the tangent-plane approximation is accurate to well below the GPS noise. What the rotation matrix actually *does* – turning a vector's global ECEF components into local East/North/Up ones – is shown in Figure 3.5.

The altitude h that enters this transform is the adopted barometric channel (Sec. 3.6.1); changing channel would move (E, N) by a relative $h/(N + h) \sim 10^{-5}$, i.e. negligibly, so the horizontal coordinates are insensitive to the altitude source. The vertical coordinate $z = U$ is therefore, up to the small deterministic tangent-plane curvature term, just the barometric height referred to the take-off value, $z \approx h - h_0$ – exactly the quantity whose dynamics were argued above to be cleanest.

3.6.6 Uniform sampling within a flight

The smoothing and differentiation below assume a *uniform* sampling interval, so each surviving flight's time base is examined next. Loggers record at a fixed nominal rate – paragliders mostly at 1 s (69 % of flights), hang gliders spread across several rates from 1 s to 10 s with a thin tail beyond (Fig. 3.7) – but the realised series can carry jitter and drop-outs. We keep each flight at its *native* rate rather than resampling to a common cadence: a common grid would either upsample slow loggers (inventing information) or throw away the resolution of fast ones, and it is unnecessary, since the derivative operator below is given the flight's own Δt (Sec. 3.6.7). What matters is uniformity *within* a flight, not across flights.

Per flight, let Δt be the median inter-fix interval. Three cases arise:

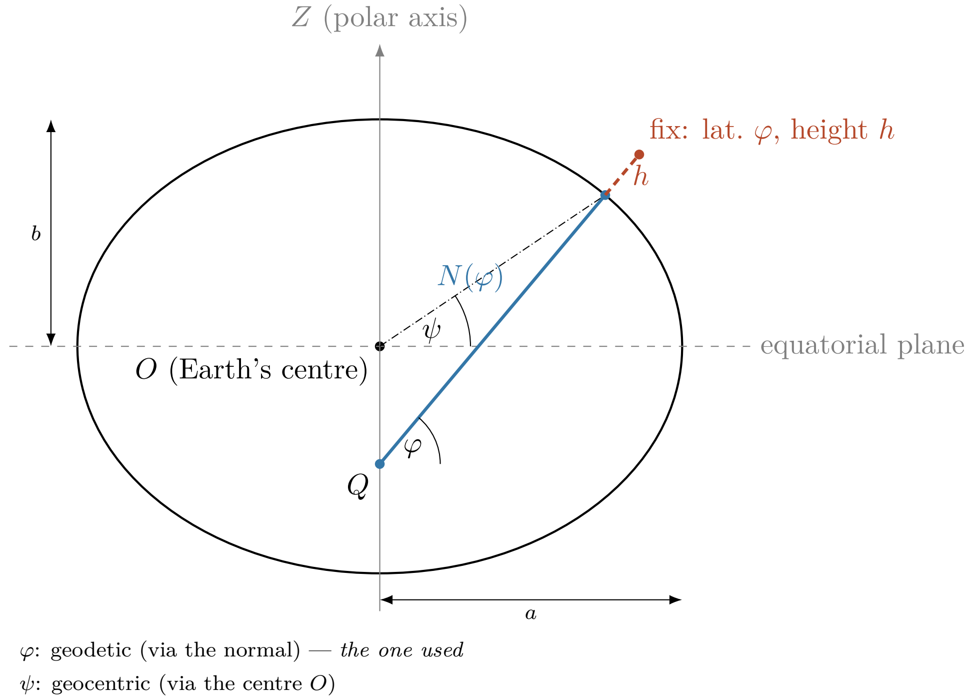

Figure 3.4: Geodetic latitude and height on the reference ellipsoid, in meridian cross-section (flattening exaggerated for visibility; true WGS84 flattening $f \approx 1/298$ would look indistinguishable from a circle at this scale): the semi-major and semi-minor axes (a, b), the geodetic latitude φ – the angle between the ellipsoid normal and the equatorial plane – and the height h above the ellipsoid along that normal. The normal does *not* pass through the centre O except at the pole and the equator; $N(\varphi)$ is precisely the distance, along the normal, from the ellipsoid surface to the point Q where it crosses the polar axis. Because of this, the geodetic latitude φ (via the normal, the one WGS84/GPS reports and the one used here) differs from the *geocentric* latitude ψ (dash-dotted, the angle of the radius O–P): $\psi < \varphi$ off the pole and equator. The third geodetic coordinate, longitude λ, is not visible in this cross-section (it is the angle *around* the polar axis, out of the page here) and is shown instead in Figure 3.5, together with how (φ, λ) orient the local frame used throughout.

Uniform. All intervals equal Δt to tolerance: the series is used as is.

Mildly irregular. Isolated jitter or single drop-outs – the largest gap is at most a few Δt and the missing fraction is small: the flight is resampled onto the uniform grid at its native Δt, interpolating across the small gaps only, and the interpolated fraction is recorded.

Broken by a long gap. A gap longer than a set bound – native, or opened where cleaning removed a frozen-lock run (Sec. 3.6.2) – is not bridged: interpolating across a long hole would fabricate dynamics the smoothing could not tell from real motion. The flight is *split* at the gap into independently analysed segments, each carried forward on its own; a segment left too short by the split is caught by the flight-level cuts (Sec. 3.6.4). A missing fraction too large even between gaps excludes the flight outright.

What is interpolated, and how. The fill is done per channel on the uniform grid. The horizontal position, which the transport analysis measures directly, is interpolated shape-preservingly (a monotone piecewise-cubic) and only ever across the short gaps a split has left behind; the barometric altitude, a smooth, low-noise, slowly-drifting sensor within a phase (Sec. 3.6.1), is interpolated linearly (or by a cubic spline) with negligible error over such gaps.

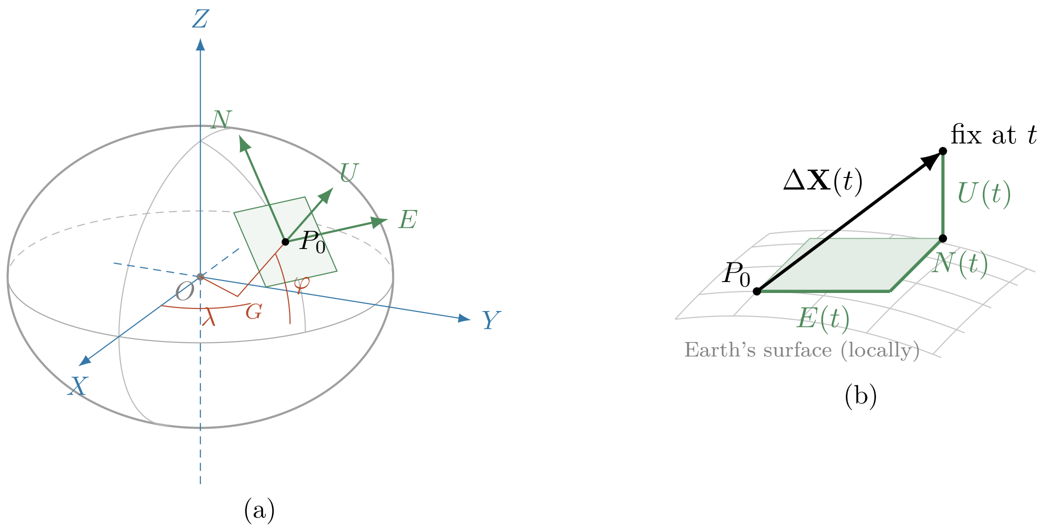

Figure 3.5: What the rotation matrix does, reproduced in our notation. (a) The global ECEF frame (X, Y, Z) in blue, centred at Earth's centre O; the two red angles that locate the take-off fix P_0 – longitude λ, measured in the equatorial plane from the prime meridian (X), and the geodetic latitude φ, the tilt of the ellipsoid normal, measured at the point G where that normal meets the equatorial plane; and the local ENU frame (E, N, U) in green, tangent to the ellipsoid at P_0. The globe is drawn as an oblate ellipsoid (flattening exaggerated, as in Figure 3.4), which makes visible that U follows the geodetic normal G–P_0 and *not* the radius from O. The green arrows are the very directions the rotation matrix sends X, Y, Z onto – its three rows, as unit vectors. (b) The same tangent plane in close-up (N receding into the page). A later fix's ECEF displacement $\Delta\mathbf{X}(t) = \mathbf{X}(t) - \mathbf{X}_0$ is the space diagonal (black) of the box whose edges are its ENU components $E(t)$, $N(t)$, $U(t)$ – exactly the three rows of the matrix above, applied to $\Delta\mathbf{X}$.

Both are filled to the *same* completeness invariant: within a retained segment every grid time carries a defined (x, y, z), so the segmentation downstream is never handed a missing value. This is where the deferred altitudes of Sec. 3.6.2 – a dropped barometric spike, a `V` fix on a GNSS-fallback flight – are actually restored.

It is tempting to reason that, because the vertical speed is about an order of magnitude smaller than the horizontal ($v_z \sim 1\,\mathrm{m\,s^{-1}}$ against $v_{xy} \sim 10\,\mathrm{m\,s^{-1}}$), the altitude tolerates *wider* gaps. It does not, and the gap bound is the same for every channel. The interpolation error is set by the *curvature* the signal has over the gap, not by its speed – a fast but straight motion is interpolated exactly, a slow but turning one is not – so a slower channel is not automatically a smoother one. More decisively, the altitude feeds v_z, the segmentation's discriminant, and a long altitude gap linearly interpolated returns a single average slope that *erases any climb–glide transition falling inside it*; and those transitions are exactly where v_z swings fastest – a thermal entry flips v_z from about -1.5 to $+3\,\mathrm{m\,s^{-1}}$ in a second or two – the worst case for interpolation. The one real asymmetry is benign and already used: an *isolated* missing altitude is filled far more reliably than a horizontal one, because the barometric channel is so smooth, which is what licenses "drop the altitude only" at cleaning – but it says nothing about widening the gap.

The gap bound is the further flight-level control promised in Sec. 3.6.4; Figure 3.6 makes the fraction of flights it leaves whole auditable in the same way as the duration and path-length cuts (implementation details: Sec. 3.5.6). One caveat carries downstream: a few observables are intrinsically Δt-dependent (a finite-difference velocity, a turning angle), so where they are compared across flights the ensemble is stratified by rate or restricted to the dominant one, rather than pretending a single cadence.

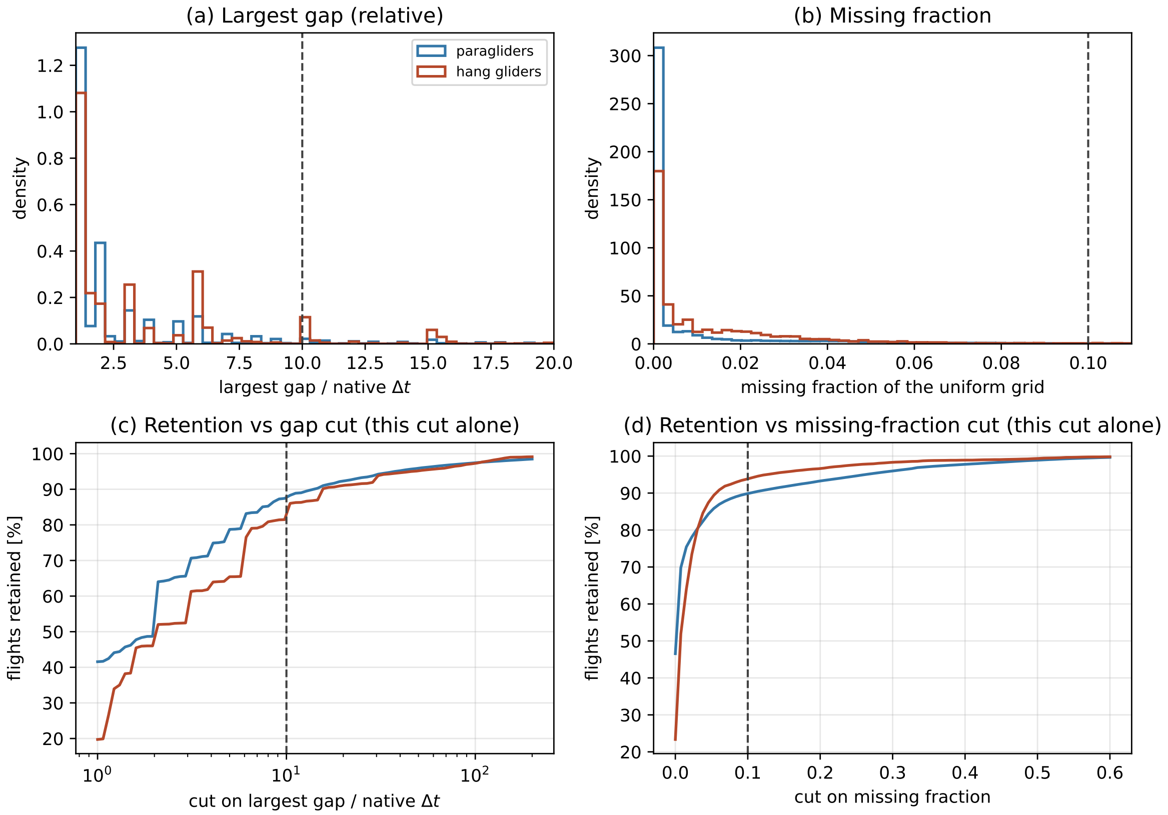

Figure 3.6: Sampling-regularity diagnostics, computed on the full dataset per discipline, on the *same* track scan as Figure 3.3 (so every parsed flight is included, not only the ones that survive the duration/path-length cuts). (a)/(b) Distribution of the largest single gap (relative to that flight's own native Δt) and of the missing fraction of a uniform grid at that interval, both on linear bins and a linear axis, over the bulk of the distribution (the heavy tail is not lost: (c)/(d) sweep the full range). Unlike native Δt (Figure 3.7) the largest-gap ratio is not purely discrete – a missed fix or two at the native rate is the single most common cause of a large gap, but the gap's exact duration is not grid-aligned to it – yet integer ratios (a gap of exactly k native intervals) are still visibly the dominant, common values, especially for hang gliders; a logarithmic axis, appropriate for a quantity spanning orders of magnitude, would have stretched precisely this small-integer structure into invisibility – it matters here because it is where the adopted cut (dashed) sits. (c)/(d) The fraction of flights each cut would retain *on its own* (marginal, not cascaded), with the adopted thresholds marked; here the cut ranges over orders of magnitude to show the full saturating shape, so (c) keeps a logarithmic axis. (implementation details: Sec. 3.5.6)

3.6.7 Smoothing and differentiation

Velocity and acceleration drive the kinematics and the segmentation, but differencing raw GPS positions amplifies the high-frequency noise characterised in Sec. 3.6.1. A **Savitzky–Golay** filter [8] solves both problems at once: it fits a low-order polynomial to a sliding window by least squares and reads off that polynomial – or its derivatives – at the window centre, returning smoothed position, velocity and acceleration (deriv $= 0, 1, 2$) in one pass per component. The derivative outputs are scaled by the flight's own Δt (Sec. 3.6.6), so the multi-rate fleet needs no common grid, and the window never crosses a flight boundary. Because it runs on the

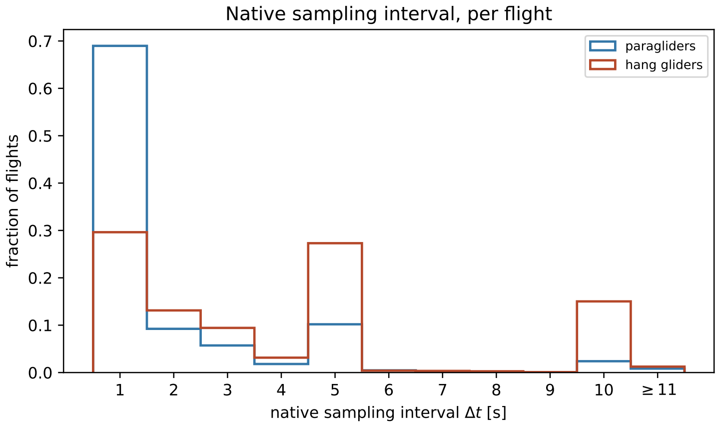

Figure 3.7: Native sampling interval Δt (median inter-fix time) per flight, on the full dataset per discipline. Δt is, in practice, a *discrete* quantity, not a continuous one: it lands on an exact whole second for the large majority of flights in both disciplines (a logger reports at one of a handful of fixed configured rates), so bins are one second wide and centred on each integer – each bar is the fraction of flights at exactly that rate, not a density (with unit-width bins the two coincide numerically, but only "fraction" describes what is actually being counted). The fleet is multi-rate: paragliders concentrate heavily at 1 s, whereas hang gliders spread across several rates, with a long thin tail beyond 11 s (last bin, aggregated). A single common cadence would waste the fast loggers; instead each flight keeps its own Δt, which the smoothing step consumes directly (Sec. 3.6.7). (implementation details: Sec. 3.5.6)

resampled grid (Sec. 3.6.6), the filter sees a uniform time base with the short gaps already filled; the interpolated points – a small, recorded fraction, since heavily-reconstructed flights are dropped whole (Sec. 3.6.2) – add a little local correlation but no bias over gaps this short.

The filter has two hyperparameters – the window length w (odd, in samples) and the polynomial order p – fixed by a *quantitative*, noise-matched procedure, not by eye:

(i) From the ensemble Welch power spectrum of the raw ENU components on a sample of flights (the diagnostic of Sec. 3.6.1 applied to E, N, U), read the frequency f_c of the knee where the dynamical signal meets the flat GPS noise floor. Its reciprocal is a smoothing timescale $\tau_c = 1/f_c$: dynamics slower than τ_c are signal, faster fluctuations are noise.

(ii) Fix p on a physical argument: to represent local curvature and give a clean second derivative the fitted polynomial must be at least quadratic, while a low order keeps the pass-band flat; $p = 3$ works and the result is insensitive to p within $\{2, 3, 4\}$.

(iii) Set the window per flight from the *timescale*, $w = \mathrm{odd}(\tau_c/\Delta t)$, so the physical smoothing scale is the same across the fleet regardless of rate.

(iv) Treat the vertical separately from the horizontal, and condition it on the altitude *source*. On a barometric flight the vertical channel is *smoother* than the horizontal – the pressure sensor carries little high-frequency noise (Sec. 3.6.1) – so it wants a shorter window, not a longer one; on a GNSS-fallback flight the vertical is instead 1.5–3× *noisier* (vertical dilution of precision) and wants more smoothing. Its knee f_c, and hence its window, are therefore read from the spectrum of the channel actually in use, separately from those of (E, N).

The choice is validated on a held-out sample: the residual (raw minus smoothed) spectrum should be flat above f_c – confirming that noise, not signal, was removed – the acceleration distribution should stay within physical bounds, and the results should be stable under a one-step change of w. Both hyperparameters are thus anchored to a measured property of the data rather than to a guess (implementation details: Sec. 3.5.7).

3.6.8 Notation

Throughout, $\mathbf{r}(t) = (x(t), y(t)) = (E(t), N(t))$ denotes the *horizontal* position in this local frame and $z(t) = U(t)$ the (barometric) altitude. By construction each trajectory starts at the horizontal origin, $\mathbf{r}(0) = \mathbf{0}$, with the clock reset to $t = 0$ at take-off, so t is the elapsed flight time (a segment created by a split at a long gap, Sec. 3.6.6, keeps its parent flight's origin and clock). The vertical is *not* re-zeroed: the take-off height is retained, so the analysis keeps both the increment that drives the dynamics (invariant to any offset) and the absolute altitude for interpretation (potential energy, convective-boundary-layer structure). The horizontal velocity is $\mathbf{v}(t) = \dot{\mathbf{r}}(t)$, with magnitude the horizontal speed $v_{xy} = |\mathbf{v}|$; the vertical velocity is $v_z = \dot{z}$. The horizontal displacement over a lag t is $\Delta\mathbf{r}(t) = \mathbf{r}(t_0 + t) - \mathbf{r}(t_0)$.

3.7 Preliminary characterization

Before the main transport analysis it is useful to run a set of lightweight checks that validate basic assumptions, reveal potential stratification needs, and characterize the dataset at the level of individual trajectories. These do not require segmentation and are conceptually simpler than the transport observables, but they are prerequisites for a well-founded analysis.

Trajectory statistics. For each flight, computed from its parsed track: airborne duration T, total horizontal path length, number of valid fixes and native sampling interval Δt. Their ensemble distributions – the filtering-relevant ones in Fig. 3.3 – characterise what the dataset contains, flag outliers, and, through Δt, set the per-flight smoothing window (Sec. 3.6.7). In the present data durations centre on a few hours and track lengths on $\sim 10^4$ fixes, so the flight-level cuts fall in an essentially empty tail.

Isotropy check. The scaling of the 1D marginal propagator (Section 4.1) requires the process to be isotropic. Competition flights have preferred directions (race goal, orography, prevailing winds), so isotropy must be tested rather than assumed.

The primary test is the **per-component variance ratio**: for an isotropic process $\langle E(t)^2 \rangle = \langle N(t)^2 \rangle = \mathrm{MSD}(t)/2$ for all t, so the ratio $\langle E(t)^2 \rangle / \langle N(t)^2 \rangle$ should be identically 1. Plotting it as a function of t reveals both the degree and the scale-dependence of any anisotropy. A complementary check is a Kolmogorov–Smirnov test comparing the empirical distributions of $E(t)$ and $N(t)$ at a few selected lags.

If isotropy fails, the two marginals $P(E(t) = x)$ and $P(N(t) = x)$ are analysed *separately*: this is both the natural fallback and, in itself, informative – unequal scaling exponents $H_E \neq H_N$ or equal exponents but different scaling functions signal geometrically anisotropic transport.

Compatibility across strata. The dataset spans more than two decades, two glider types (paragliders and hang gliders), and a range of performance classes (Sec. 3.3) – from recreational EN A/B paraglider wings to CCC racing wings and the flexible/rigid hang-glider classes – whose kinematics differ substantially; sailplanes will later add a third type. Before treating all flights as draws from a single statistical process – i.e. before computing ensemble averages over the pooled dataset – the subpopulations must be checked for statistical compatibility.

A lightweight way to do this: for each stratum (e.g. a single wing class, or the flights of a single season) compute the per-component displacement variance $\langle E(t)^2 \rangle$ as a function of the elapsed time t and overlay the curves on one plot. If they coincide, the transport statistics are compatible and pooling is justified. If one stratum grows systematically faster or with a different shape – competition wings, say, cover ground faster than recreational ones – the strata must be analysed separately, and that difference is itself a result about the universality of the transport regime.

Spatial coverage. A map of take-off coordinates and displacement end-points at fixed t reveals geographic clustering (France has distinct orographic zones: Alps, Pyrenees, Massif Central, plains). If flights from mountainous terrain differ markedly from flatland flights, geographic stratification may be needed alongside temporal and aeronautical stratification.

Chapter 4

Next steps

With the dataset assembled, cleaned and characterised (Chapter 3), the analysis proceeds as follows.

4.1 Observables on the whole trajectory

A first group of observables requires *no* segmentation: it can be measured directly on the full trajectories and already probes the transport at the global level. For each, we state what it is, how it is computed, and what it is for.

Transport and diffusion.

Mean-squared displacement (MSD). With every flight starting at the origin ($\mathbf{r}(0) = \mathbf{0}$), the *ensemble-averaged* MSD is the mean squared horizontal distance reached after an elapsed time t,

$$\mathrm{MSD}(t) = \langle\, |\mathbf{r}(t)|^2 \rangle, \tag{4.1}$$

where $\langle \cdot \rangle$ is an average *over flights* (an ensemble average) at fixed t. Its growth law $\mathrm{MSD}(t) \sim t^\alpha$ fixes the transport regime ($\alpha = 1$ Brownian, $\alpha > 1$ super-diffusive, $\alpha < 1$ sub-diffusive; equivalently the Hurst exponent $H = \alpha/2$). This is the primary test for anomalous transport.

Time-averaged MSD (TA-MSD). A second MSD estimator is built entirely *within* a single flight. For flight k of duration T_k, one slides a window of width t along the trajectory $\mathbf{r}^{(k)}$ and averages the squared displacement over all starting times t_0 *inside that flight*:

$$\overline{\delta^2}^{(k)}(t) \;\equiv\; \frac{1}{T_k - t} \int_0^{T_k - t} \left|\mathbf{r}^{(k)}(t_0 + t) - \mathbf{r}^{(k)}(t_0)\right|^2 \mathrm{d}t_0. \tag{4.2}$$

The average over t_0 here is a *time* average internal to flight k; it has nothing to do with the ensemble average $\langle \cdots \rangle$ over different flights used in the MSD. Because a single-flight curve $\overline{\delta^2}^{(k)}(t)$ is noisy, the operationally useful quantity is its mean over all flights of comparable duration T:

$$\mathrm{TAMSD}(t, T) \;\equiv\; \left\langle \overline{\delta^2}^{(k)}(t) \right\rangle_{T_k \approx T}. \tag{4.3}$$

Ergodic case. For an ergodic process, $\mathrm{TAMSD}(t, T) \to \mathrm{MSD}(t)$ as $T \to \infty$, independently of T, and the individual values $\overline{\delta^2}^{(k)}(t)$ concentrate around the common limit (the distribution collapses to a delta function).

Non-ergodic case: two independent signatures [9].

(i) *Aging.* $\mathrm{TAMSD}(t,T)$ differs from $\mathrm{MSD}(t)$ and retains a systematic dependence on T: flights of different duration yield different curves. This is a strong indicator of non-ergodicity, but it is sufficient, not necessary – some non-ergodic processes show only a weak T-dependence.

(ii) *Non-self-averaging.* The distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories remains broad even as $T \to \infty$: individual flights give systematically different TA-MSD values and the distribution does not collapse. This is the more fundamental signature: it persists even when aging is weak.

We therefore monitor both $\mathrm{TAMSD}(t,T)$ and the full distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories at fixed t and T. The canonical illustration – a CTRW with power-law waiting times – is worked out analytically in Appendix A, where both EA-MSD≠TA-MSD and the T-dependence are derived explicitly.

Propagator. Since each flight starts at the origin, $\mathbf{r}(0) = \mathbf{0}$, the position at time t and the displacement from the start coincide: $\mathbf{r}(t) = \mathbf{r}(t) - \mathbf{r}(0)$. We therefore define the propagator as the probability density of the position vector $\mathbf{x} \in \mathbb{R}^2$ at time t,

$$P(\mathbf{x},t)\,\mathrm{d}^2x = \Pr\bigl(\mathbf{r}(t) \in [\mathbf{x},\, \mathbf{x} + \mathrm{d}\mathbf{x}]\bigr). \tag{4.4}$$

For a *self-similar* process with exponent H the propagator obeys the scaling form

$$P(\mathbf{x},t) = t^{-2H}\, G\!\left(\frac{\mathbf{x}}{t^H}\right), \tag{4.5}$$

where $G : \mathbb{R}^2 \to \mathbb{R}_{>0}$ is a time-independent scaling function (normalised, $\int G\,\mathrm{d}^2u = 1$).

In practice one works with the **one-dimensional marginal** obtained by projecting onto a single Cartesian component x,

$$P(x,t) \equiv \int_{-\infty}^{+\infty} P((x,y),t)\,\mathrm{d}y. \tag{4.6}$$

Assuming the process is *isotropic*, $G(\mathbf{u}) = G(|\mathbf{u}|)$, the integral can be evaluated by the substitution $y = t^H v$:

$$P(x,t) = t^{-2H} \int_{-\infty}^{+\infty} G\!\left(\sqrt{\left(\tfrac{x}{t^H}\right)^2 + v^2}\right) t^H\,\mathrm{d}v = t^{-H}\, F\!\left(\frac{x}{t^H}\right), \tag{4.7}$$

where $F(u) \equiv \int_{-\infty}^{+\infty} G(\sqrt{u^2+v^2})\,\mathrm{d}v$ inherits the normalisation $\int F\,\mathrm{d}u = 1$ from G. The marginal therefore satisfies the same scaling form as the 2D propagator, with the same exponent H but in one dimension. The shape of F distinguishes transport regimes: a Gaussian F characterises ordinary diffusion, while heavy power-law or stretched-exponential tails signal anomalous transport.

The marginal $P(x,t)$ is the object actually estimated from data: use the East component, the North component, or both (if isotropy holds, they coincide and can be pooled; if not, they are analysed separately – see Section 3.7). The scaling collapse consists in plotting $t^H P(x,t)$ against x/t^H for several values of t: all curves should fall on the single function F.

Probability of return to the origin. The value of the propagator at its centre, $P_0(t) \equiv P(\mathbf{0},t)$, i.e. the height of the peak. For a self-similar process it decays as a power law set by the same exponent, $P_0(t) \sim t^{-dH}$ in d dimensions (t^{-2H} in the plane, t^{-H} for a single component). It therefore yields H from the *bulk* of the distribution rather than from its tails, which is valuable when heavy tails make the variance – and hence the MSD – ill-defined; a change of slope in $\log P_0$ versus $\log t$ exposes a crossover between regimes, and a peak exponent that disagrees with the MSD exponent is itself informative, pointing to multi-scaling. (This is the route Mantegna and Stanley [10] used to read off the index of Lévy-like walks.)

Non-Gaussianity. How far the propagator departs from a Gaussian, measured by the non-Gaussian parameter (in the plane)

$$\alpha_2(t) = \frac{\langle |\mathbf{r}(t)|^4 \rangle}{2\left\langle |\mathbf{r}(t)|^2 \right\rangle^2} - 1, \tag{4.8}$$

which vanishes for a Gaussian and is positive for heavier-than-Gaussian tails (the factor 2 is the two-dimensional Gaussian value of $\langle|\mathbf{r}|^4\rangle/\langle|\mathbf{r}|^2\rangle^2$). Evaluated on the propagator $P(\mathbf{x}, t)$ at each elapsed time t, it is a function of t – one number per time – tracing a curve $\alpha_2(t)$ that pinpoints the scales over which the displacement statistics are non-Gaussian (typically largest at intermediate times, where occasional long glides dominate, and decaying if a central-limit Gaussianisation sets in at long times). It summarises the shape of P, it does not replace it.

Velocity autocorrelation $C(\tau)$. $C(\tau) = \langle \mathbf{v}(t_0) \cdot \mathbf{v}(t_0 + \tau)\rangle$ (often normalised by $C(0)$) is the dot product of the velocity with its later self: it is large and positive when, a delay τ later, the motion still points the same way, so the scale over which $C(\tau)$ decays *is* the memory time of the heading. It controls the MSD through the Green–Kubo relation (for a stationary velocity) $\mathrm{MSD}(t) = 2\int_0^t (t - \tau)\, C(\tau)\, \mathrm{d}\tau$: a slowly decaying C produces a faster-growing MSD. We do not presume the motion ends in ordinary diffusion; rather, the functional form of $C(\tau)$ is the diagnostic – an integrable C relaxes to normal diffusion, a non-integrable (e.g. power-law) tail sustains persistent super-diffusion with no crossover, and a sign change reveals the anti-persistence expected from circling.

Turning-angle distribution. The distribution of the angle θ between two successive horizontal displacement vectors along the trajectory – the change of heading from one step to the next. A distribution peaked at $\theta = 0$ means strong forward persistence (a correlated random walk); a flat distribution means memoryless reorientation. This is the microscopic origin of the persistence seen in $C(\tau)$ and the MSD.

First-passage and first-exit times. The first-passage time $T(L)$ is the first time the horizontal distance from take-off reaches a value L (the first-exit time, the first time the trajectory leaves a disc of radius L) [11]. For a self-similar transport process the characteristic time scales with the length as $\langle T(L)\rangle \sim L^{1/H}$, with dynamic exponent $z = 1/H$, so the slope of $\log\langle T\rangle$ against $\log L$ gives an estimate of H obtained in the space domain, independent of the MSD; the full distribution of $T(L)$ additionally exposes heavy tails coming from long trapping events.

Speed and vertical-velocity distributions. The magnitude of the horizontal velocity and the vertical velocity. Basic kinematic characterization; the vertical velocity also separates climbing from sinking and feeds the segmentation below.

Trajectory geometry.

Tortuosity. How much a path winds, quantified by the ratio between the length actually travelled and the straight-line distance between its endpoints over the same interval (1 for a straight line, larger for a more convoluted path). It is a geometric *efficiency* descriptor and does not map one-to-one onto a transport exponent – very different dynamics can share a tortuosity – so we use it descriptively and to separate phases (close to 1 in glides, large in search and climb) rather than as an estimator of H.

Fractal dimension. The box-counting dimension d_f of the trail, i.e. the exponent with which the number of boxes of size ϵ needed to cover the path grows as $\epsilon \to 0$. For a self-similar walk it is tied to the transport exponent by

$$d_f = \min(1/H,\, 2), \tag{4.9}$$

so a ballistic line ($H = 1$) gives $d_f = 1$, a Brownian path ($H = 1/2$) gives $d_f = 2$, and the reported sub-ballistic soaring ($H \approx 0.88$) gives $d_f \approx 1.14$. The fractal dimension is thus the geometric counterpart of H and a third, independent route to it, after the MSD slope and the return/first-passage scalings; agreement among the three is a strong internal check, disagreement is evidence of multi-scaling.

Radius of gyration. The root-mean-square distance of the trajectory points from their centroid, i.e. the characteristic spatial extent of a flight. As a single number it sets the natural length scale of a flight (useful to normalise other observables); measured as it grows with elapsed time, $R_g(t) \sim t^H$, it offers yet another reading of the same exponent.

One number per flight, or a function of scale? The geometric quantities above can be read in two complementary ways. As a *single number per flight* – one tortuosity, one d_f, one R_g – whose distribution over the ensemble characterises the population and feeds the stratified comparisons. Or as a *function of scale or elapsed time* – $R_g(t)$ as the trajectory unfolds, tortuosity measured on sub-segments of growing duration t, the box count as a function of box size – which is precisely where the transport exponents live ($R_g(t) \sim t^H$; the box-count slope is d_f). The scalar is descriptive; the scale dependence is what links geometry back to the anomalous exponent, and both views will be used.

4.2 Segmentation into flight phases

Building on the previous work of Vilpellet et al. [4], each flight is segmented into the three phases of the soaring cycle:

Transition (glide). The inter-thermal cruise: a near-straight, energy-efficient glide toward the next source of lift (sinking, large directed horizontal displacement). This is the directed "step".

Search. The exploratory phase entered when no thermal is found at a comfortable altitude: gentle turns and probing manoeuvres to sample the air mass and locate the next thermal while limiting altitude loss. It mediates the passage from one step to the next trap.

Climb (thermalling). Circling within rising air to regain altitude: persistent turning with small net horizontal displacement. This is the localized "trap".

That work performs the segmentation with a hidden Markov model [12] whose observations are a small set of *binary* trajectory-derived features — the sign of the vertical speed, the persistence of the curvature angle, and indicators of trajectory straightness.

A first task of this thesis is to scrutinize this segmentation rather than take it for granted: to assess whether those binary variables are adequate and robust on the larger, more heterogeneous dataset assembled here, and whether the feature set or the model should be refined, extended, or replaced. The segmentation method is therefore itself an object of study, and the specific variables are deliberately left open at this stage. This three-state description refines the two-state caricature underlying continuous-time random walks and Lévy walks.

4.3 Phase-resolved observables

The segmentation unlocks a second group of observables, defined per phase occurrence and then as ensemble distributions. These are the building blocks of an intermittent-transport description and, unlike those of Section 4.1, cannot be obtained without first segmenting the flight.

The waiting phase: climb and search. Between two consecutive glides the flier advances little horizontally: it searches for a thermal and climbs in it. For a transport model these two phases together form the *waiting time* between directed steps, so we treat them on an equal footing and build three distributions – the climb duration τ_c, the search duration τ_s, and their sum $\tau_w = \tau_s + \tau_c$, the total time spent essentially stationary between glides. The distinction is not cosmetic: Vilpellet et al. [4] report the search times to be broadly *power-law* distributed while the climb times are closer to *exponential*, so that the heavy tail of the combined waiting time $\psi(\tau_w)$ – the quantity that actually decides whether trapping drives sub-diffusion – would be inherited from the search phase. Whether this still holds on the larger dataset is itself one of the things to re-check once the segmentation is in place. We therefore report $\psi(\tau_w)$ together with the separate $\psi(\tau_c)$ and $\psi(\tau_s)$. Alongside the durations we record the climb kinematics (altitude gained Δz, mean climb rate, thermalling radius and period) and the search geometry (path length, tortuosity, net displacement) – the search being the phase in which prior work locates the signature of pilot skill.

The directed step: glide (transition). Each glide contributes a horizontal step of length ℓ and duration τ_g, with ground speed, glide ratio (horizontal distance per unit altitude lost) and sink rate. Because a glide is close to ballistic, $\ell \approx v_g \tau_g$, step length and duration are expected to be nearly proportional; we test this directly through the joint law of (ℓ, τ_g) – the ℓ–τ_g scatter, the conditional mean $\langle \ell \mid \tau_g \rangle$, and the spread of the ratio ℓ/τ_g (the glide speed). Both marginals matter: the tail of the step-length distribution $\lambda(\ell)$ controls super-diffusive contributions, so we compute the distributions of ℓ and τ_g separately as well as their joint law.

Anatomy of a flight. Number of thermals (climbs) per flight, inter-thermal distances, and the fraction of both time and along-track distance spent in each phase: the coarse composition that sets the weight of each phase in the global transport.

Angular diffusion between glides. The turning angle between the heading of one transition segment and the next – the segment-level counterpart of the turning angle of Section 4.1. How fast successive glide directions decorrelate is the key ingredient of the minimal model below.

Two couplings then decide which transport model applies, and are stated here as explicit objects of study. The first is the *step–waiting coupling*: whether a long glide tends to be followed by a long wait, i.e. whether ℓ and τ_w are correlated. A genuine coupling is the defining feature of a Lévy walk and is what separates it from a decoupled CTRW, in which jumps and waiting times are drawn independently.

The second is the *memory in the phase sequence*. A flight is a discrete sequence over the three states $\{\text{transition}, \text{search}, \text{climb}\}$, and memory can hide in two places. (i) The *order* of the phases, summarised by the transition matrix $P_{ij} = \Pr(\text{next} = j \mid \text{current} = i)$: a perfectly cyclic flight (T $\to$ S $\to$ C $\to$ T) gives a near-deterministic matrix, whereas aborted climbs, repeated searches or skipped phases appear as off-cycle probabilities and reveal how stereotyped the soaring cycle actually is. (ii) The *durations*, through correlations between consecutive phase durations – for instance whether a long climb tends to precede a long glide. Both probe the *renewal* assumption underlying the textbook CTRW and Lévy-walk results, in which each step and wait is independent of the history: if the transition matrix is structured or the durations are correlated, the process is non-renewal and those formulas must be amended. Measuring these is therefore what tells us whether a memoryless model is admissible at all.

4.4 Reproducing the reference results

A first objective of the analysis, undertaken once the segmentation is in place, is to check whether the enlarged—and eventually multi-source—dataset reproduces the quantitative and graphical results of Vilpellet et al. [4]. That study covers paragliders, hang gliders and sailplanes; the present data cover the first two (Chapter 3), so both the paraglider and the hang-glider comparisons are within reach, while the sailplane figures become reachable once that source is added. Two figures are the main targets.

The first is the per-phase conditional MSD (their Fig. 3): the transition segments should stay ballistic ($\alpha = 2$, i.e. $H = 1$) across time lags; the search phase should be strongly sub-diffusive ($H \approx 0.3$ for paragliders and hang gliders, ≈ 0.2 for sailplanes); and the climb phase should be quasi-ballistic at long lags but with an effective velocity about two orders of magnitude smaller, including the transient anti-correlation near the $\sim 30\,\mathrm{s}$ thermalling period.

The second is the set of per-phase descriptive statistics (their Fig. 4): the glide ratios (≈ 8.5, 10, and 25 for paragliders, hang gliders, and sailplanes), the horizontal-velocity statistics, the heavy-tailed distribution of transition times, the search-time fraction and the effective search radius, the vertical climb velocities, the thermalling radii, and the thin, near-exponential distribution of climb times—together with the global time budget ($\approx 60\%$ transition, $\approx 35\%$ climb, the remainder searching) and the global sub-ballistic scaling ($H \approx 0.88$).

Recovering these results would validate the acquisition pipeline and the segmentation and would test the universality of the reported behaviour on a substantially larger sample; systematic departures would be equally informative, pointing to dataset- or method-dependent effects to investigate. This comparison is therefore the bridge between the present data-acquisition stage and the original analysis that follows.

4.5 Transport analysis

Using these observables, test for anomalous transport: estimate the MSD scaling exponent α, fit the tails of $\psi(\tau)$ and $\lambda(\ell)$ with principled methods (e.g. maximum-likelihood power-law fits with goodness-of-fit tests [13]), and perform model selection between the continuous-time random walk and the Lévy-walk frameworks [1,2] – using the step/waiting-time coupling to tell them apart. Because logging devices, sites, gliders and weather differ, results will be stratified (by season, site, glider class, and data source) and trajectories normalized, and finite-size, censoring and sampling biases will be controlled throughout.

4.6 Modelling and simulation

The aim is a *minimal* stochastic model that reproduces the asymptotic anomalous scaling reported for soaring (a Hurst exponent $H \approx 0.88$) with as few ingredients as possible. A preliminary working hypothesis—to be made precise, and possibly revised, once the empirical phase statistics are available—is that the transport is carried mainly by the transition (glide) phases. The model would therefore resolve the transition segments explicitly while merging search and climb into a single effective non-transition state, so that the soaring cycle reduces to a sequence of directed glides interrupted by reorientations. Anomalous transport would then emerge from the directed transition segments combined with an *angular diffusion* of the heading from one transition to the next: successive glide directions decorrelate gradually rather than abruptly, producing a correlated random walk whose persistence controls the long-time exponent. Simulated ensembles would be compared with the data to test whether these ingredients suffice. These are baseline ideas, to be refined—or changed—as the analysis progresses.

4.7 Tooling

Mirror the acquisition package with an analysis package, and let the figures and tables of this document be generated automatically from the analysis outputs – as the dataset statistics already are – so that this document stays in step with the work and remains a reliable basis for the thesis and for presentations.

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Appendix A

CTRW with power-law waiting times: EA-MSD vs TA-MSD

This appendix derives the ensemble-averaged MSD (EA-MSD) and the time-averaged MSD (TA-MSD) for the continuous-time random walk (CTRW) with heavy-tailed waiting times, and shows explicitly that the two differ and that the TA-MSD depends on the observation time T.

Model

A walker takes steps of length l_i (iid, symmetric, $\langle l^2 \rangle = \sigma^2 < \infty$) separated by waiting times τ_i (iid) drawn from

$$\psi(\tau) \sim \frac{\alpha\, \tau_0^\alpha}{\tau^{1+\alpha}}, \qquad \tau \to \infty, \quad 0 < \alpha < 1. \tag{A.1}$$

The mean waiting time diverges ($\langle \tau \rangle = \infty$), so the walker is anomalously slow. Let $n(t)$ be the number of steps completed by time t, $T_n = \sum_{i=1}^n \tau_i$ the time of the n-th step, and

$$x(t) = \sum_{i=1}^{n(t)} l_i \tag{A.2}$$

the position at time t.

Ensemble-averaged MSD

Since steps and waiting times are independent,

$$\langle x^2(t) \rangle = \sigma^2 \, \langle n(t) \rangle. \tag{A.3}$$

For large t the mean number of renewals in $[0, t]$ follows from the renewal equation and the Tauberian theorem applied to the Laplace transform of ψ. Because $\hat{\psi}(s) \equiv \int_0^\infty e^{-s\tau} \psi(\tau)\, \mathrm{d}\tau \simeq 1 - (\tau_0 s)^\alpha + \cdots$ for $s \to 0$ (a consequence of ψ having a power-law tail with exponent $1 + \alpha$), the renewal theorem gives

$$\langle n(t) \rangle \sim \frac{t^\alpha}{\tau_0^\alpha\, \Gamma(1+\alpha)}, \qquad t \to \infty. \tag{A.4}$$

Hence the **EA-MSD** grows sub-linearly,

$$\mathrm{MSD}(t) = \langle x^2(t) \rangle \sim \frac{\sigma^2}{\tau_0^\alpha\, \Gamma(1+\alpha)}\, t^\alpha \equiv K_\alpha\, t^\alpha, \qquad 0 < \alpha < 1. \tag{A.5}$$

The transport is sub-diffusive with Hurst exponent $H = \alpha/2 < 1/2$.

Time-averaged MSD and its mean

For a trajectory of duration T, the time-averaged MSD at lag $t \ll T$ is

$$\overline{\delta^2}(t;T) = \frac{1}{T-t}\int_0^{T-t}\big[x(t_0+t)-x(t_0)\big]^2\,\mathrm{d}t_0. \tag{A.6}$$

Since $\langle[x(t_0+t)-x(t_0)]^2\rangle = \sigma^2\langle n(t_0+t)-n(t_0)\rangle$, the mean of the integrand equals σ^2 times the mean number of steps in any window of width t. The key observation is that the integral $\int_0^{T-t}[n(t_0+t)-n(t_0)]\,\mathrm{d}t_0$ counts, for each step k that lands at time $T_k \in [0,T]$, the length of the t-wide sliding window that contains T_k. For T_k deep in the interior ($t \le T_k \le T-t$) this length is exactly t; the contributions from the two boundary layers of width t are negligible for $t \ll T$. Therefore

$$\int_0^{T-t}\big[n(t_0+t)-n(t_0)\big]\,\mathrm{d}t_0 \approx t\,n(T), \qquad t \ll T, \tag{A.7}$$

and

$$\overline{\delta^2}(t;T) \approx \frac{\sigma^2\,t\,n(T)}{T}. \tag{A.8}$$

Taking the ensemble average and using $\langle n(T)\rangle \sim K_\alpha T^\alpha/\sigma^2$,

$$\mathrm{TAMSD}(t,T) \equiv \langle\overline{\delta^2}(t;T)\rangle \approx \sigma^2\,\frac{\langle n(T)\rangle}{T}\,t \sim K_\alpha\,T^{\alpha-1}\,t. \tag{A.9}$$

Comparing (A.5) and (A.9):

- The EA-MSD scales as t^α (sub-diffusive), whereas the TAMSD scales as t^1 (linear in the lag) – a qualitatively different dependence on t.
- The TAMSD carries an explicit $T^{\alpha-1}$ prefactor: since $\alpha < 1$ this is a decreasing function of T, so longer flights yield smaller TAMSD at the same lag. This T-dependence is the signature of *aging*: the process has not forgotten its past, and the answer depends on how long we have observed it.
- The ratio $\mathrm{TAMSD}(t,T)/\mathrm{MSD}(t) \sim (t/T)^{1-\alpha} \to 0$ as $T \to \infty$ at fixed t – the two observables diverge.

Distribution of the TA-MSD: non-self-averaging

Equation (A.8) shows that $\overline{\delta^2}(t;T) \approx \sigma^2 t\,n(T)/T$, so the randomness in the TA-MSD comes entirely from $n(T)$, the total number of steps in $[0,T]$. For a power-law renewal process the rescaled count

$$\xi \equiv \frac{n(T)}{\langle n(T)\rangle} \tag{A.10}$$

converges in distribution as $T \to \infty$ to a non-degenerate limit law: the *Mittag-Leffler distribution* M_α, whose probability density function is defined through its moments $\langle\xi^n\rangle = n!\,\Gamma(1+\alpha)^n/\Gamma(1+n\alpha)$ [9]. This is a broad, non-Gaussian distribution on $(0,\infty)$ that does *not* concentrate around $\xi = 1$ for $\alpha < 1$: different trajectories give systematically different values of $\overline{\delta^2}(t;T)$ even in the limit $T \to \infty$. This is *non-self-averaging*: the TA-MSD of a single trajectory is not a reliable estimate of any ensemble quantity, regardless of how long the trajectory is.

Quantitatively, the ergodicity-breaking parameter

$$\mathrm{EB}(t,T) \equiv \frac{\langle[\overline{\delta^2}(t;T)]^2\rangle - \langle\overline{\delta^2}(t;T)\rangle^2}{\langle\overline{\delta^2}(t;T)\rangle^2} = \mathrm{Var}(\xi) \sim c_\alpha\,T^{-\alpha} \to 0 \tag{A.11}$$

does tend to zero (the variance of ξ decreases with T), but the *distribution* of ξ converges to a non-trivial limit rather than to a delta function. This is why monitoring the full distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories – not just its mean – is the proper diagnostic for non-ergodicity: the collapse (or non-collapse) of that distribution as T grows tells us whether we are dealing with a self-averaging ergodic process or a non-self-averaging non-ergodic one.

For comparison, in an ordinary random walk ($\alpha = 1$, finite mean waiting time) the ratio $n(T)/\langle n(T)\rangle$ tends to 1 in probability as $T \to \infty$, so the distribution of the TA-MSD collapses to a delta function, $\overline{\delta^2}(t;T) \to \mathrm{MSD}(t)$ almost surely, and the process is ergodic.

Appendix B

Power spectral density and Welch's method

This appendix explains the tool used in Chapter 3 to compare the two altitude channels of an IGC file, namely the *power spectral density* (PSD) estimated with *Welch's method* [14]. It is written so the reader need not already know what these are.

What the power spectral density is

Take a signal sampled at a fixed rate, here an altitude series $z_n = z(n\,\Delta t)$, $n = 0, \ldots, N-1$, with sampling interval Δt (so the sampling frequency is $f_s = 1/\Delta t$). Any such finite signal can be written as a sum of sinusoids of different frequencies (its discrete Fourier transform). The **power spectral density** answers a simple question: *how is the "power" (the variance) of the signal distributed across frequencies?* A slow drift puts its power at low frequency; rapid jitter puts its power at high frequency. Formally the PSD $S(f)$ is the Fourier transform of the autocovariance of the signal, and it is normalised so that the total area under it equals the variance of the signal,

$$\int_0^{f_s/2} S(f)\,\mathrm{d}f \;=\; \mathrm{Var}(z), \tag{B.1}$$

with units of (metres)^2 per hertz. The upper limit $f_s/2$ is the *Nyquist frequency*: a series sampled every Δt cannot resolve anything faster than one oscillation per two samples. Reading a PSD is therefore reading *where in frequency* the signal lives: this is exactly what separates a slowly drifting but locally clean channel (power concentrated at low f) from a channel with a high-frequency noise floor (power that stays large up to the Nyquist frequency).

The periodogram and why it is noisy

The naive estimator of $S(f)$ is the **periodogram**: take the discrete Fourier transform $\hat{z}(f) = \sum_n z_n\, e^{-2\pi i f n \Delta t}$ and form $|\hat{z}(f)|^2$ (suitably normalised). It is asymptotically unbiased but *not consistent*: its variance does not shrink as the record gets longer – a longer signal gives more frequency points, each still as noisy as before. Plotted directly it is a spiky, unreadable curve, from which one cannot reliably compare two channels.

Welch's method: segment, window, average

Welch's method turns the noisy periodogram into a usable estimate by *averaging.* The recipe is:

(i) **Segment.** Split the record into several shorter, overlapping segments.

(ii) **Window.** Multiply each segment by a smooth taper (a Hann window) that falls to zero at its ends. This suppresses the spurious high-frequency power (*spectral leakage*) that an abrupt segment boundary would otherwise inject.

(iii) **Average.** Compute the periodogram of each windowed segment and average them.

Averaging K segments cuts the variance of the estimate by roughly a factor K, at the cost of coarser frequency resolution (set by the segment length rather than the whole record). The result is a smooth PSD estimate whose *level* can be compared meaningfully between two signals – the standard, robust spectral estimator. Each segment is typically detrended before transforming – e.g. by removing a linear fit – so that the estimate reflects the fluctuations about the trend – the part where sensor noise lives – rather than the trend itself (implementation details: Sec. 3.5.1).

How these spectra read for the two altitude channels – and what the comparison implies for the choice of channel – is discussed where that choice is made (Sec. 3.6.1), not here; this appendix covers only the method.

Implementation and Computational Details

Chapter 1

Overview

This part collects the implementation- and computation-level detail behind the choices made in the main text: the exact code module and script that produced each figure or table, the configuration values in force, sample sizes and computation times, and the status of each piece (implemented and tested, designed but not yet built, or planned). It is organised as a document within the document: its own chapters mirror the body chapter by chapter, so that a detail removed from the body sits in the corresponding chapter or section here. It is not meant to be read start to end – the body links to the relevant place at the point where a detail was trimmed out of the text, via a cross-reference such as "(implementation details: Sec. 3.5.2)".

Contents of this part

Table 1.1 is a quick index of which script reproduces which auto-generated figure or table (all paths relative to `scripts/reporting/`); the chapters that follow go into the reasoning and numbers behind each one.

Figure/Table	Script	Notes
Stats macros; Tables 3.1, 3.2	`generate_stats.py`	Reads committed `data/*/seasons_index.csv`; instant; part of the pre-commit hook.
Census macros (`\StatScan...`)	`generate_census_stats.py`	Reads the cached track scans on the SSD (never rescans); instant; best-effort in the pre-commit hook, so the committed macros refresh whenever the SSD is mounted.
Figure 3.1	`generate_altitude_noise_figure.py`	Needs the raw IGC archive (env. `SOARING_PARA_DATA_ROOT`/`SOARING_DELTA_DATA_ROOT`).
Figures 3.2, 3.3, 3.6, 3.7	`generate_preproc_figure.py`	One run produces all four (a shared full-census scan feeds three; fix-level cleaning uses a separate sampled pass). Needs the raw IGC archive.

Table 1.1: Script that regenerates each auto-generated thesis figure or table. Run as `uv run python scripts/reporting/<name>`; only `generate_stats.py` is wired into `scripts/build_docs.sh` (the two figure scripts need the external SSD and are run manually).

Chapter 2

Data acquisition

Mirrors Chapter 2.

2.1 Pipeline and tooling

Package `soaring.acquisition.ffvl`, exposed as two CLI entry points, `soaring-para` and `soaring-delta`, sharing subcommands: `fetch-xml` (archive season XMLs), `download` (fetch IGC files), `build-catalog` (rebuild catalog/index), `status` (per-season coverage), `verify` (post-hoc integrity scan, Sec. 2.3).

Resumable downloads. Files already present on disk are skipped, so an interrupted job resumes freely without re-fetching.

Atomic writes. Each file is written to a temporary `.part` and then renamed, so a file under its final name is always complete – important on the exFAT external disk, which has no journaling.

Content validation. A response is accepted only if it looks like a real IGC file (an `A` header record followed by at least one `B` fix record); HTML error pages and truncated responses are rejected and retried.

Courteous networking. A small bounded thread pool, per-request timeouts, exponential backoff, and a minimum delay between requests avoid hammering the server; a browser TLS fingerprint is impersonated to pass the site's Cloudflare challenge.

2.2 Reproducibility

Destination directory (`data_root`) set per machine via an environment variable – `SOARING_PARA_DATA_ROOT` for paragliders, `SOARING_DELTA_DATA_ROOT` for hang gliders – or a config file shipping a placeholder (`configs/para_download.yaml`/ `configs/delta_download.yaml`); every command validates the setting at start-up.

2.3 Integrity

The `verify` subcommand re-scans every downloaded file on disk, reusing the same IGC validation applied at download time, and flags truncated or invalid files, leftover `.part` fragments, and read errors.

Chapter 3

The dataset

Mirrors Chapter 3, including its pre-processing and preliminary- characterization sections (Secs. 3.6, 3.7) – the cleaning and first-look steps are part of describing the dataset, not of the planned analysis in Chapter 4.

3.1 The IGC tracklog format

The parser (`soaring.analysis.igc.parse_igc`) checks, at decode time and independently of any flight-dynamics judgement: that the time is a valid 24-hour `hh:mm:ss`; that the arc-minutes field of latitude and longitude is `< 60` (the `DDMM.mmm` encoding is undefined otherwise); that the hemisphere letter is one of the two valid characters; and that the decoded latitude/longitude fall within $[-90, 90]/[-180, 180]$ degrees. A record failing any of these is dropped as unparseable, on the same footing as a record with the wrong length or a non-numeric field. This format-validity check is implemented now and covered by unit tests; the fix-level dynamics-plausibility checks it is distinct from (Sec. 3.5.2) are designed and thresholded but not yet implemented.

3.2 Flight metadata and the catalog

Both `catalog.csv` and `seasons_index.csv` are regenerated from the archived season XMLs on demand, via the acquisition CLI's `build-catalog` subcommand (Sec. 2.1) – never hand-edited.

3.3 Glider categories

Label normalisation (not yet implemented) will merge `aile_class` variants and recover a blank or placeholder ("0") class from the wing model (`aile`) where possible, at the fix-level/pre-processing stage (Sec. 3.5.2).

3.4 Coverage and statistics

To be populated: season-table provenance.

3.5 Trajectory pre-processing

Mirrors Sec. 3.6 and its subsections below. The Welch/PSD computational procedure used to compare altitude channels (Sec. 3.5.1) is the same one Sec. 3.6.7 will reuse, applied to E, N, U instead of altitude alone; 3.5.7 should cross-reference 3.5.1 for that shared procedure rather than restate it, once populated.

3.5.1 Choice of altitude channel

PSD sample (Figure 3.1a–c). A seeded random subsample of 3,000 flights per discipline – larger than a spectral *shape* alone would need, because the high-frequency barometric curve is visibly under-averaged (ragged) at a few hundred flights and smooths out only with a few thousand. Panel (a)/(b) use one representative flight (the one with the most fixes in the subsample); its measured offset (panel b) is annotated on the panel itself, not restated here.

Welch/PSD computational procedure (panel c). IGC loggers record one fix per second nominally, but timestamps are not guaranteed to be perfectly regular (a missed fix, a logger writing at 1.05 s instead of 1.00 s); Welch's method (Appendix B) requires a uniformly sampled signal, so the following runs on every flight before the transform.

(i) **Modal sampling period.** For each flight the median inter-fix interval is computed and rounded to the nearest integer second. The value that appears most often across all flights in the subsample – the *mode* – is taken as the common target Δt (in practice $\Delta t = 1\,\mathrm{s}$ for both disciplines, giving $f_s = 1\,\mathrm{Hz}$ and $f_{\mathrm{Nyq}} = 0.5\,\mathrm{Hz}$).

(ii) **Flight selection.** Only flights whose own median period lies within 0.25 s of the modal Δt are retained, so pooled spectra share one frequency axis; flights with fewer than $N_{\mathrm{seg}} = 256$ fixes ($\sim$4 min at $\Delta t = 1\,\mathrm{s}$) are also excluded, since at least one full segment is needed. Of the 3,000 sampled flights per discipline, this leaves 1,641 paragliders but 727 hang gliders: paragliders are predominantly 1 s loggers (69 %, Sec. 3.5.6), so the population mode is a good match for most of them, whereas hang gliders split across several native rates (1, 2, 5, $\geq$8 s, Figure 3.7) and only their 1 s-logger minority survives this cut.

(iii) **Uniform resampling.** The altitude series $z(t)$ is linearly interpolated onto a grid $t_0, t_0 + \Delta t, t_0 + 2\Delta t, \ldots$ spanning the flight, without introducing frequency content beyond f_{Nyq}.

(iv) **Welch on the resampled series.** `scipy.signal.welch` is called with $f_s = 1/\Delta t$, $N_{\mathrm{seg}} = 256$, 50% overlap, a Hann window, and linear detrending per segment.

(v) **Robust ensemble estimator.** The per-flight PSDs are kept individually (per discipline and channel) and reduced across flights by the per-frequency *median*, with a shaded 10 to 90 percentile band; panel (c) then pools the two disciplines, whose medians coincide. The median, not the arithmetic mean, because these per-flight spectra are heavy-tailed: a mean is dominated by a few flights and swings by one to two orders of magnitude between samples – for the hang-glider barometric floor, from 0.8 to $93\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$ between an $n = 114$ and an $n = 727$ draw – whereas the median is stable. Because Δt is the same for every retained flight by construction, the figure's Nyquist dotted line is a single fixed value.

Reading the band (panel c). The two channels' medians coincide because, for the typical flight, both are dominated at high frequency by the metre resolution to which the IGC format rounds every altitude: the white-noise floor of a 1 m quantiser, $\frac{1}{12}/f_{\mathrm{Nyq}} \approx 0.17\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$, sits just below where the measured medians level off ($\approx 0.3\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$). The channels part only in the *tail*: at the 90th percentile the high-frequency GNSS floor is 4× the barometric one for paragliders and 108× for hang gliders – the worse GNSS multipath of the hang-glider fleet, and the reason its GNSS band is the wider – and 13 % to 20 % of flights carry a GNSS floor above three times their own barometric one. That heterogeneity, not a uniform gap, is what motivates preferring the uniformly-clean barometric channel (Sec. 3.6.1).

Fallback-rate census (panel d). A full census, not a sample: the same full parse of every track is needed anyway for the flight-level filtering diagnostics of Sec. 3.5.4 (Figure 3.3), so the barometric-presence fraction is read off that one scan rather than measured separately – exactness is a byproduct, not extra work. (A sized random sample – the standard proportion formula $n = z^2 p(1-p)/e^2$ – would be the right tool if a full census were not already being paid for elsewhere; it is not needed here.)

Presence is whole-file (panel d). The barometric channel is all-or-nothing at the file level, checked on the same census: when it is absent it reads exactly zero for 99.7 % (paragliders) / 100 % (hang gliders) of those flights, and when present it covers every fix for 96 % / 98 % of them (the rest miss only a handful of individual fixes, dropped one by one, Sec. 3.6.2). A logger either has a pressure sensor for the whole track or none at all, so the `alt_source` decision is per flight, never per fix.

3.5.2 Fix-level cleaning

This section gives the working definitions the narrative (Sec. 3.6.2) states in words: the three detectors, their parameters, the order they run in, and how the whole step is validated. The three families of Table 3.3 map onto three detectors of increasing context: an absolute bound (no context), a robust local test (a short window of context), and a structural rule (a whole run).

Status. The format-validity rejection is implemented in `soaring.analysis.igc.parse_igc` and covered by unit tests. The dynamics and structural cleaning below is designed, and its absolute bounds are thresholded on the data (Table 3.3, Figure 3.2); the routine that walks a trajectory applying all three detectors, and the robust-test and frozen-lock parameters it needs, are the next step. The parameter names introduced here are the config keys that step will add.

Robust local-outlier test (off-the-trend defects). For a fix P_k at time t_k, take the fixes lying within a window of $\pm w$ seconds of t_k (a *time* window, so it adapts to the native cadence and to irregular sampling), excluding P_k itself. Fit a robust local estimate of the position there – the component-wise median of the windowed fixes, or a degree-one Theil–Sen fit – and evaluate it at t_k; the residual r_k is the great-circle distance from P_k to that estimate. Scale it by a robust local spread $\sigma_k = 1.4826$ MAD of the windowed residuals (the factor makes σ_k a Gaussian-consistent standard deviation [15]), and flag P_k when

$$r_k > k\,\sigma_k \quad \text{and} \quad r_k > \varepsilon_{\min}.$$

This is a Hampel identifier [16]: a median centre, a MAD scale, a threshold in robust standard deviations. Working defaults, to be finalized by the injected-defect test below: $k \approx 4$–5, a window w of a few native intervals (order 15 s to 30 s), and an absolute floor $\varepsilon_{\min} \approx 15$ m, a few times the horizontal GPS noise.

Three points earn the test its place over the bare speed bound. *The floor $\varepsilon_{\min}$ is not cosmetic*: on a very smooth stretch $\sigma_k \to 0$, so without it any sub-metre wiggle would exceed $k\sigma_k$ and the test would chase GPS noise where there is no outlier. *It is local, and locality buys attribution*: a speed belongs to the segment *between* two fixes and cannot say which endpoint is at fault, whereas the residual belongs to one fix – when the absolute bound does fire, the identifier names the culprit, and a burst is removed as one block. The flag itself is context-sensitive – a $30\,\mathrm{m\,s^{-1}}$ step out of a $12\,\mathrm{m\,s^{-1}}$ glide is flagged while the same step passes unremarked where it is ordinary – but a flag alone does not delete (next paragraph): such a flagged-but-physically-possible fix is *kept*, and the flag survives as a per-flight anomaly count, recorded alongside the integrity-gate counters as a diagnostic. *It does not clip real manoeuvres*: a genuine sharp turn is drawn by consecutive, mutually consistent fixes, each well predicted by its neighbours and so of small residual; only an isolated point off the local trend scores large. The one failure mode – a turn so tight it is under-sampled at the native cadence – is rare at these rates, and the injected-defect test below measures it directly. A short burst of consecutive spikes inflates the windowed MAD only mildly, so it is still flagged, and is removed as the block whose bracketing neighbours rejoin within scale. The identical test runs on the horizontal position and, independently, on the adopted altitude channel, so a vertical spike invalidates the altitude alone and the horizontal position stays (the channel decoupling of Sec. 3.6.2).

From flag to action: detect here, reconstruct later. The identifier only *flags*; it never repairs. A flagged horizontal fix is *deleted* as a node – not overwritten with the local median – so that all reconstruction is done by the single resampling step (Sec. 3.5.6), under one policy and one accounting, rather than by a second, in-place imputation that would then feed the resampler as if it were data. This is the *Hampel identifier* [16], the detection half of the eponymous filter, not the median-substituting filter itself: the local median/Theil–Sen enters only as the reference for the residual r_k, never as an output value. And a flag alone does not delete: to be removed, a fix must be flagged *and* corroborated as physically impossible – its implied in-and-out speed exceeding the absolute v_{xy} bound – so that the one benign false positive, a genuine under-sampled tight turn (whose flagged fix still moves at ordinary flying speed), is flagged but *kept.* The gate deliberately narrows the identifier's *deletions* to what the absolute bound already deems impossible: on its own, the local test attributes and audits. A locally anomalous but physically possible fix is kept by design – its residual harm is bounded and absorbed by the smoother, whereas deleting it would risk real dynamics – the same asymmetric-risk argument that shapes the whole step. A flagged altitude is treated more permissively, marked missing on the flag alone, because a dropped altitude is not a deletion but a deferral: it is restored by the single audited interpolation at resampling, so the stakes are far lower than for a position node. This is the fix-level face of the *default to keep* rule of Sec. 3.6.2: removal needs positive, corroborated evidence.

Frozen-lock detection (on-the-trend). A run is a lost-lock episode only under the rule of Eq. (3.3) – spatially collapsed, *and* either barometrically flat *or* explicitly declared, *and* long enough – and the pass defaults to keeping the data otherwise. (i) *Extent*: the run's bounding diameter – the maximum great-circle distance between *any* two of its fixes, not the fix-to-fix step – stays below ε (a few times the GPS noise, $\sim$10 m to 15 m); a circling climb spans its disc, $2R \approx 40\,\mathrm{m}$ to $100\,\mathrm{m} \gg \varepsilon$, and is cleared on extent alone, whereas a step-based test would fire on every 1 Hz climb (step $v_{xy}\Delta t \approx 10\,\mathrm{m} \sim \varepsilon$). (ii) *Independent witness*: over the run the barometric altitude must also be flat – $|\Delta z_{\mathrm{baro}}|$ below a few metres – since a real thermalling climb gains tens of metres in that time on a sensor independent of the GNSS, so a horizontally-still run whose baro altitude is still climbing is a tight climb and is kept. (iii) *Corroboration*: a declared lost lock (the `V` flag, GNSS altitude `00000`) or byte-identical repeated coordinates confirm it – and, being an *or*-branch of Eq. (3.3), override the witness: a run of byte-identical coordinates under a climbing barometric altitude is a freeze that happened *during* a climb (no real circling wing rewrites the same coordinates), and is cut – the positions are invented even where the height gain is real. The run must also span at least τ_{freeze}, set to a few thermalling periods ($\gtrsim 60\,\mathrm{s}$ to $90\,\mathrm{s}$), so that a long, slow, tight climb is never mistaken for a freeze. Only then is the run marked as a data gap; the split it induces is realized at the uniform-sampling stage (Sec. 3.5.6), on the same footing as a native long gap. On a

GNSS-fallback flight the barometric witness is unavailable, so there the extent and flag tests must both hold outright and, in any doubt, the run is kept – the safe direction, since keeping a rare true freeze costs far less than splitting a real climb.

Duplicate and backward timestamps (on-the-trend). The parser already rebuilds a monotone clock: it unwraps the UTC-midnight roll-over (a drop of ∼86,400 s advances the day count) and clamps residual backward jitter with a running maximum (Sec. 3.1). The redesign changes one behaviour: a backward step that is *not* a roll-over is no longer silently clamped but removed – the fix with the corrupt time is deleted as a node, just as a fix with a corrupt position is. A falsified timestamp is a defect to record, not a repair to hide. This requires moving the clamp out of the parser: `parse_igc` keeps the roll-over unwrap (a format concern, needing the raw time of day) but stops flattening backward jitter with the running maximum – flattened, the evidence the cleaning must act on no longer exists downstream. Fixes sharing one UTC second are collapsed to a single representative at that second's centroid position. This replaces the provisional "keep the first" of the earlier design: keeping the first privileges an arbitrary member and, for a genuine ≥ 2 Hz logger, would silently halve the cadence, whereas the centroid is order-independent and the sub-second resolution it discards is discarded anyway once the flight is resampled to a common Δt (Sec. 3.5.6). One dilution case is accepted: if one member of a duplicate pair is itself a position spike, the centroid halves its amplitude before the outlier pass (which runs after the time defects) sees it; a large spike is still flagged at half size, and one small enough to slip under $k\sigma_k$ after halving is within what the smoother absorbs.

Absolute bounds (out-of-range, the backstop). The six numbers of Table 3.3 are the `FixLevelThresholds` dataclass, loaded by `soaring.analysis.preprocessing.load_preproc_config` from `configs/preprocessing.yaml` (key `fix_level`); nothing is hard-coded in Python. They are the context-free floor of the design, and are placed on the real per-fix distributions (Figure 3.2) rather than guessed. The horizontal-speed bound is set where the bin-to-bin ratio of a finely binned (2.5 m s^{-1}) histogram settles at ≈ 1 – where the decay has actually stopped – not where it first visibly slows (≈ 40 m s^{-1} for hang gliders, versus the adopted 55 m s^{-1}), since a shallow but real decreasing stretch can still be genuine rare dynamics.

Diagnostic sample (Figure 3.2). Computed by `fix_level_distributions` and rendered by `make_fixlevel_diagnostics_figure` (Table 1.1). Panels (a)–(c) pool a seeded random sample of 15,000 paraglider flights (126,723,499 consecutive-fix pairs, ∼73 s to compute) and the complete 6,716-flight hang-glider population (a full census at this size, 37,915,334 pairs, ∼25 s) – under two minutes total. The large sample is deliberate: resolving the sparse artefact tail well enough to tell it from real dynamics needs far more points than a headline statistic would. The x-range in (a)/(b) extends to each discipline's own 99.99th percentile rather than cropping just past the bound; panel (c) keeps a narrower range, its cut sitting far out in an otherwise tight, unimodal distribution. Panel (b) bins one integer m s^{-1} per bin because barometric altitude is logged to the metre: differencing at a 1 s cadence returns integer rates, and a finer grid would alias them into a spurious comb. (Not every flight samples at 1 s, Figure 3.7, and differencing over a longer native step returns finer values – multiples of 0.2 m s^{-1} at 5 s – that partly fill the gaps once cadences are pooled: among 1 s-native fixes about 99% land within 0.05 m s^{-1} of an integer, against only 34–42% at every other cadence.)

Order, and a single pass. The detectors run in a fixed order – format, then the time defects (backward, duplicate), then position outliers, then frozen-lock, then altitude – because a speed or a residual is only meaningful once the time base is monotone and free of duplicates. The position-outlier and frozen-lock passes are one left-to-right scan that carries the last accepted fix as an anchor: it never re-walks the whole track, and it absorbs a run by extending the anchor across the flagged block – tracking the block's running bounding box to test its extent against ε – so the cost is $O(N)$ in the number of fixes.

Flight-level integrity gate. The pass records how much it changed each flight (or segment) – `n_fix_raw`, `n_fix_clean` and `frac_interpolated` in `flights_meta`. A flight whose deleted-plus-reconstructed fraction exceeds a small f (order 0.05 to 0.10) is dropped whole: beyond that point the trajectory is more interpolation than measurement. This is the cleaning counterpart of the sampling-regularity cut (Sec. 3.5.6).

Configuration. The absolute bounds are already the `FixLevelThresholds` group above. The robust-test parameters (w, k, $\varepsilon_{\min}$), the frozen-lock parameters (the extent tolerance ε, the minimum span τ_{freeze} and the barometric-flatness threshold of the witness test) and the integrity fraction f join that group in `configs/preprocessing.yaml` when the routine is built; as with every threshold in this pipeline they live in the config, never in Python.

Validating the cleaning. The removed-fraction count of Figure 3.2 bounds only the false positives (over-cleaning). Two further checks, part of the same planned step, bound the false negatives and the bias. *Injected defects*: on tracks the cleaning leaves untouched, plant synthetic defects of known type and size – position spikes of amplitude a, frozen runs of length τ, backward-time glitches – and measure the recovered fraction against size. This gives the detection floor (the smallest spike caught) and calibrates k, w, ε and τ_{freeze} to a target rate, the only direct measurement of what the cleaning misses. *Downstream invariance*: recompute the transport estimators – the displacement-scaling exponent and the waiting-time-tail exponent – over a grid of cleaning strengths, and take the operating point on the plateau where they stop moving; movement under stricter cleaning is the signature of a cut biting into signal. A cheap third check guards against a missed defect class: the fixes flagged by the absolute bound should nest inside those flagged by the local test (the impossible is a subset of the locally anomalous), so a large disagreement points to a defect neither was designed for. A fourth check targets the error this pipeline most wants to avoid – losing a slow-phase fix. Once the segmentation is in place (Ch. 4) the removal and split rates are measured *per phase* and required to be flat across transition, search and climb rather than enriched in the two slow phases; and synthetic tight climbs – circles at flying speed, over a range of radii and wind drift – planted in otherwise clean tracks confirm directly that the frozen-lock detector never splits a real climb. This is the false-*positive* complement of the injected-defect test, and its natural readout is the tail of $\psi(\tau_w)$: a cut biting into climb or search would thin that tail before it moved anything else. A fifth check targets the vertical bound specifically: the fixes cut by $|v_z| > 13\,\mathrm{m\,s^{-1}}$ must be *isolated*, not consecutive – a sustained spiral dive holds $15\,\mathrm{m\,s^{-1}}$ to $20\,\mathrm{m\,s^{-1}}$ of real sink, so a run-shaped excess is the signature of a genuine manoeuvre, calling for a raised bound or a coherent-run exemption rather than censoring; an isolated super-threshold difference, bracketed by ordinary rates, is a sensor error and safe to cut.

Reproduce. Table 1.1, row `generate_preproc_figure.py` (one run also regenerates Figures 3.3, 3.6 and 3.7 from a shared full-census scan; the fix-level panels use a separate sampled pass).

3.5.3 Trimming of ground phases

The take-off speed ($5\,\mathrm{m\,s^{-1}}$) and sustained duration (30 s) are the `TrimmingThresholds` dataclass, loaded from `configs/preprocessing.yaml` (key `trimming`); as with every threshold in this pipeline, nothing is hard-coded in Python.

3.5.4 Flight-level filtering

The duration ($T \geq 40\,\mathrm{min}$) and path-length ($L \geq 30\,\mathrm{km}$) cuts are the `FlightLevelThresholds` dataclass, from `configs/preprocessing.yaml` (key `flight_level`).

Diagnostic sample (Figure 3.3). Computed by `load_or_scan_tracks` (via `scan_tracks` and `track_stats`), rendered by `make_flightlevel_diagnostics_figure`, on the full census – every parsed track, no sub-sampling: 186,025 paraglider and 6,716 hang-glider tracks. (The totals – like every census number in this document – are generated macros quoting the cached scan (`generate_census_stats.py`); the live archive grows by a trickle, so they can trail the download counts of the season tables – 186,052 paragliders at the last stats regeneration – until the cache is refreshed.) Median duration and path length are 2.6 h/86 km for paragliders and 3.1 h/158 km for hang gliders – the latter flying faster and farther. This same scan (cached at `<data_root>/derived/track_scan.parquet`) also feeds Figures 3.6 and 3.7, so all three share one full-census pass.

Reproduce. Table 1.1, row `generate_preproc_figure.py`.

3.5.5 From geographic to local Cartesian coordinates

No implementation content: this step is pure coordinate geometry with no configurable parameters, and both of its figures (Figure 3.4, Figure 3.5) are hand-drawn TikZ diagrams, not data plots.

3.5.6 Uniform sampling within a flight

The gap bound (10× native Δt) and the maximum missing fraction (0.10) are loaded from `configs/preprocessing.yaml` (key `sampling`) as the `SamplingThresholds` dataclass. The bound is on the *time base* and is the same for every channel: an order-of-magnitude smaller v_z buys no wider gap for the altitude, since the interpolation error scales with curvature, not speed, and a wide altitude gap would erase the climb–glide transitions that v_z is there to resolve (Sec. 3.6.6).

Per-channel fill. Across a short gap the horizontal position is interpolated shape-preservingly (monotone piecewise-cubic, e.g. `scipy.interpolate.PchipInterpolator`) and the barometric altitude linearly or by cubic spline (its within-phase smoothness makes the choice immaterial). The result meets the completeness invariant of Sec. 3.6.2: within each retained segment every grid time carries a defined (x, y, z) – deleted position nodes and deferred (missing) altitudes both restored here – so the segmentation is never handed a hole; long gaps are split, not filled, and `frac_interpolated` records the reconstructed fraction per segment for the integrity gate.

Figures 3.6 and 3.7. Both rendered by `make_gap_diagnostics_figure`/`make_sampling_figure` from the same full-census scan as Figure 3.3 (Table 1.1, row `generate_preproc_figure.py`) – no separate parse needed.

Gap-ratio discreteness (Figure 3.6). Checked directly: only 86 %/63 % (para/hang) of largest-gap-ratio values are exact integers, the rest a genuinely continuous scatter; the integer peaks visible for hang gliders sit around $k = 3, 6, 10, 15$.

Native-rate discreteness (Figure 3.7). Checked directly: Δt lands on an exact whole second for 99.9 % of paragliders and 100 % of hang gliders. Per-rate breakdown: paragliders mostly (69 %) at 1 s; hang gliders spread across 1 s (30 %), 5 s (27 %), 10 s (15 %), 2 s (13 %), and a long thin tail beyond 11 s.

3.5.7 Smoothing and differentiation

Not yet implemented. The polynomial order ($p = 3$) and the three smoothing timescales – `tau_c_horizontal_s` for (E, N), `tau_c_vertical_baro_s` and `tau_c_vertical_gnss_s` for U – are the `SavgolParams` dataclass in `configs/preprocessing.yaml` (key `savgol`), but today hold placeholder values – each τ_c is to be fixed once the ENU power spectra are measured, reusing the Welch procedure of Sec. 3.5.1 applied to E, N, U instead of altitude. The vertical timescale is read from the spectrum of the altitude channel *actually in use*: a barometric flight, being the smoother channel, takes a shorter window than (E, N), a GNSS-fallback flight (noisier by vertical dilution of precision) a longer one – which is why the config carries the conditioning on `alt_source` as two explicit vertical keys rather than one value (Sec. 3.6.7). The filter runs on the resampled grid, so the small interpolated fraction (`frac_interpolated`) is already in place when it reads its window.

3.6 Preliminary characterization

Not yet implemented; see Sec. 3.7.

Chapter 4

Next steps

The sections below mirror Chapter 4, which – since pre-processing and preliminary characterization now live in Chapter 3's mirror (Sec. 3) – starts from "Observables on the whole trajectory." None of this analysis code exists yet, so each entry is a placeholder rather than a populated record; each will gain its own implementation content (module names, sample sizes, reproduction commands) as that code is written.

4.1 Observables on the whole trajectory

Not yet implemented; see Sec. 4.1.

4.2 Segmentation into flight phases

Not yet implemented; see Sec. 4.2.

4.3 Phase-resolved observables

Not yet implemented; see Sec. 4.3.

4.4 Reproducing the reference results

Not yet implemented; see Sec. 4.4.

4.5 Transport analysis

Not yet implemented; see Sec. 4.5.

4.6 Modelling and simulation

Not yet implemented; see Sec. 4.6.

4.7 Tooling

Largely underway already: see Sec. 4.7 and the reproducibility map, Table 1.1.